Introduction

Two types of signal

This demonstrates the need for "Advanced" Transforms

Stationary:

The frequency content is the same for all time. ⇒ contain a limited amount of information ⇒ be of limited use.

Non-stationary:

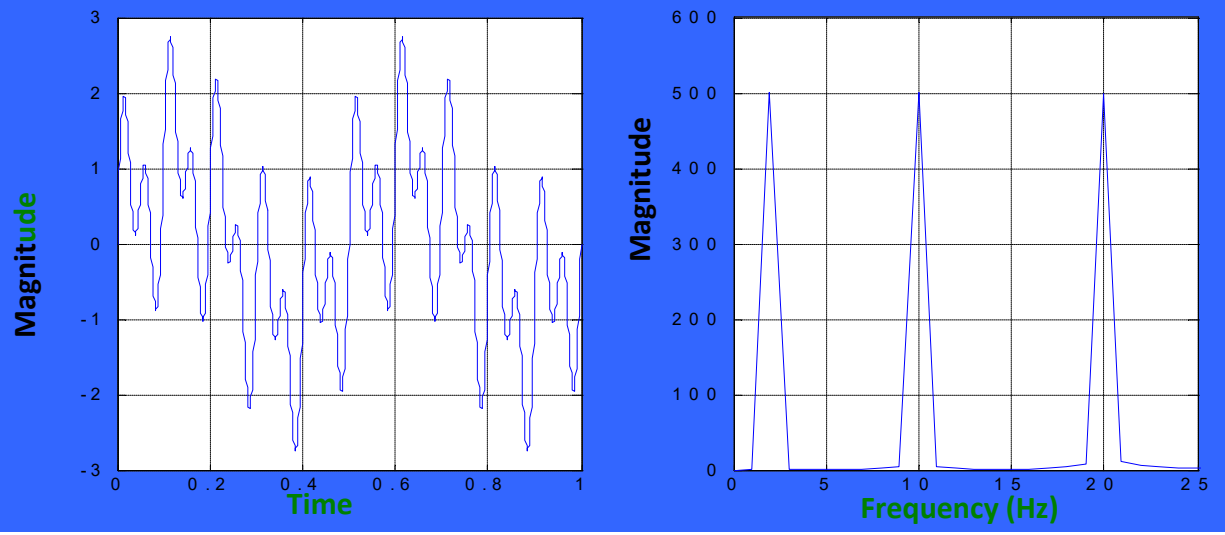
The frequency content changes with time.

Example 1: Stationary signal

平稳信号

2 Hz + 10 Hz + 20Hz

Stationary



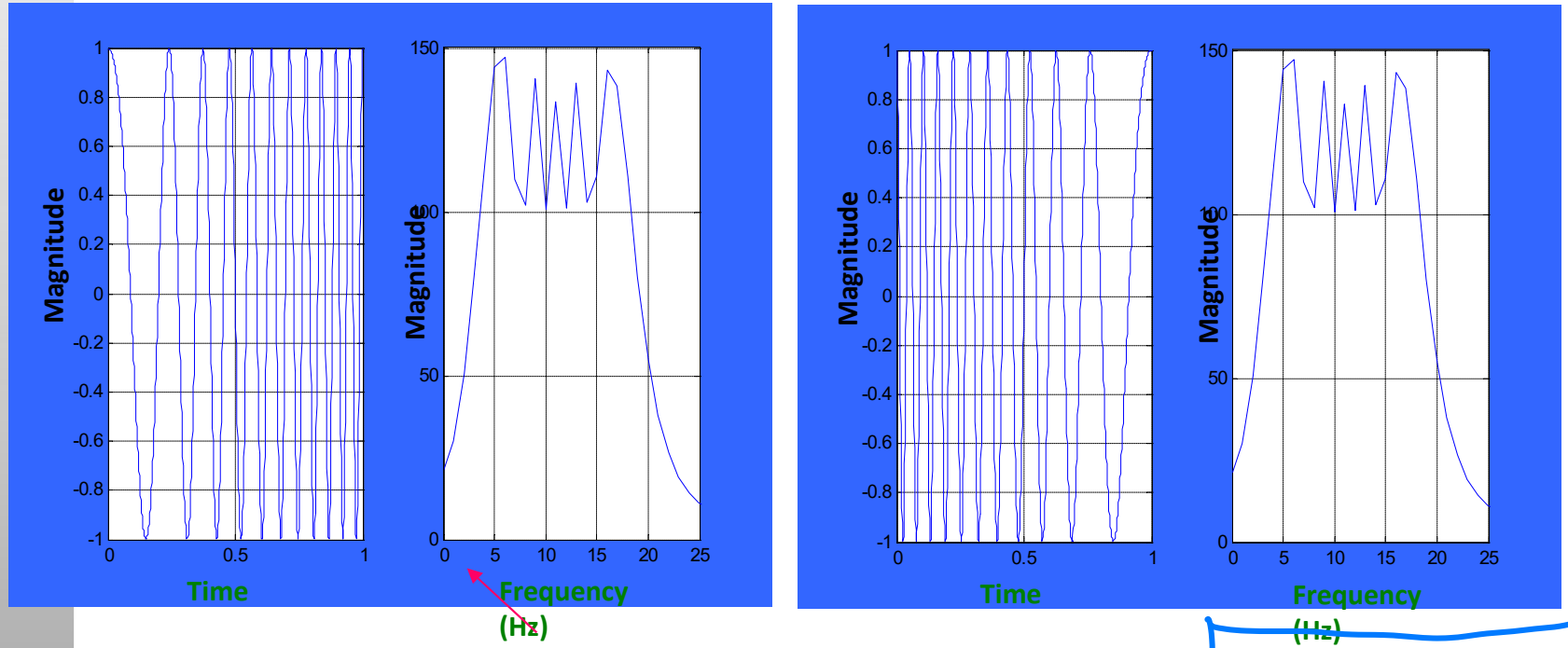
Signal in time domain Power spectrum $|X(f)|^2$

By looking at the Power spectrum (Fourier Transform) of the signal we can recognize the three frequency components (at 2,10,20Hz respectively).

Example 2: Non-stationary signals

非平稳信号

Consider two linearly modulated sinusoids (chirps).
The first with increasing frequency and the second with decreasing frequency. with time



In this case we have two nonstationary signals in time with identical FTs.
Confusion arises and power spectrum (Fourier Transform) is not useful.

Summary: Fourier Transform

Kind of Decomposition	Frequency
Analyzing Function	Sines and cosines, oscillating indefinitely
Variable	Frequency
Information	The frequencies that make up the signal
Suited for	Stationary signals
Notes	With the FFT, it takes *n*log*n* computations to compute the Fourier transform of *n* points

$N \log_2 N$

Summary: Windowed Fourier Transform (Short time Fourier Analysis)

Kind of Decomposition	Time-Frequency
Analyzing Function	Wave limited in time, multiplied by sinusoidal oscillations. Window size is fixed, but the frequency inside the window varies.
Variable	Frequency, position of the window
Information	The smaller the window, the better the time information. The cost is low frequency information. Large windows give better frequency information but less timing precision.
Suited for	Quasi-stationary signals (stationary at scale of window)
Notes	When a Gaussian is used as the window envelope this is the Gabor transform. Though the Fourier transform is orthogonal, in general the windowed transform is not.

Summary: Wavelet Transform

Kind of Decom-position	Time-Scale
Analyzing Function	Wave limited in time with fixed number of oscillations. Wavelet is contracted or dilated to change the window size and change the scale at which one looks at the signal. Since number of oscillations is fixed, frequency of the wavelet changes as scale changes.
Variable	Scale, position of the wavelet
Information	Small wavelets provide good time information but poor frequency information. Vice versa for large wavelets.
Suited for	Nonstationary signals, such as brief signals and signals with components at different time scales.
Notes	Wavelet transform can be continuous or discrete. Orthogonal and biorthogonal wavelets are special discrete cases. With orthogonal wavelets, the transform can be computed with *cn* computations. *C* depends on the complexity of the wavelet.

Summary: Wigner-Ville Distribution

Kind of Decomposition	Time-Frequency
Analyzing Function	Uses the signal itself. Motivated by time-frequency energy density (c.f. a probability density).
Variable	Time and Frequency. Has high resolution in both time and frequency.
Suited for	Simple signals, e.g. linear chirp, gaussian pulse
Notes	More complex signals lead to undesired “cross-terms”. Can be suppressed with smoothing, but lose high resolution in the process.

Linear Algebra

Inner Product ("dot product") 内积

For two (real) vectors $\mathbf{a}$ and $\mathbf{b}$ of same dimension n (长度)

inner product is $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_i a_i b_i$

Sometimes write as $\mathbf{a} \cdot \mathbf{b}$ ("dot product") or as $\mathbf{a}^t \mathbf{b}$

Example : given $\mathbf{a} = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$ we get

$$\mathbf{a}^t \mathbf{b} = \begin{bmatrix} 2 & 5 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix} = 2 \times 4 + 5 \times 3 + 1 \times 5 = 28$$

- *Matlab:* » `a'*b`

Orthogonal and Orthonormal Vectors

Vectors **a** and **b** are *orthogonal* to each other if

$\mathbf{a}^t\mathbf{b} = 0$ i.e. their inner product $\langle \mathbf{a}, \mathbf{b} \rangle$ is zero.

Vectors are *orthonormal* if orthogonal and of unit length.

A *set* of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$, [i.e. $\{\mathbf{v}_i\}$ with $i = 1, 2, ..., m$]

is orthonormal if

$$\mathbf{v}_i^t \mathbf{v}_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{otherwise} \end{cases}$$

This is called the Kronecker Delta Function

Example: The 3-dimensional basis system given by

$$\mathbf{v}_1 = [1\ 0\ 0],\ \mathbf{v}_2 = [0\ 1\ 0],\ \mathbf{v}_3 = [0\ 0\ 1]$$

is orthonormal.(For example, x, y, z axes). Dot products are taken 2 at a time.

For *n*-dimensional vectors, we can have no more than *n* vectors in any set of orthogonal vectors.

Explain why it is desirable for a transform to have orthogonal basis functions, and explain what needs to be done if they are not orthogonal.

Any vector in a vector space can be represented by a combination of other vectors in that space [1 mark]. If this set of vectors for the vector space is orthogonal, then any other vector in that space can be represented by a unique set of coefficients [1 mark]. This set of vectors then forms a basis function [1 mark]. This is then used to perform the transform (1 mark). The same set of orthogonal vectors can then be used to invert the transform [1 mark]
if they are not orthogonal, then another set of vectors that are mutually orthogonal to the first must be found to allow the inverse transform [1 mark]. These two sets of vectors are referred to as a biorthogonal set [1 mark]

Basis vectors

A set of vectors $\{\mathbf{w}_i\}$ $i = 1,2,...,n$ is said to be

a *basis* for a vector space R^n if the set of vectors $\{\mathbf{w}_i\}$

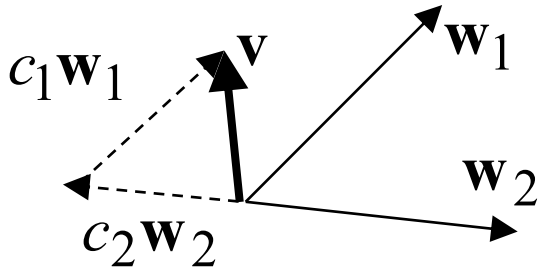
(a) is linearly independent, and

(b) spans the space R^n.

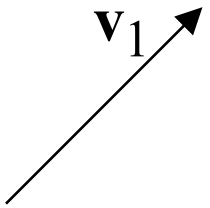
There must be exactly n vectors in the basis.

Each vector $\mathbf{v}$ in R^n has a unique set of *coordinates* $\{c_i\}$ in the basis $\{\mathbf{w}_i\}$, where $\mathbf{v} = c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + \cdots + c_n\mathbf{w}_n$

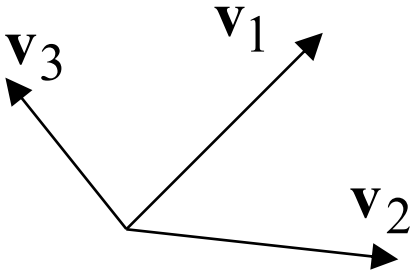
Set $\{\mathbf{w}_1, \mathbf{w}_2\}$ is a basis for R^2



Not a basis for R^2 (Doesn't span R^2)



Not a basis for R^2 (linearly dependent)



Row Space & Column Space; Rank

The column space $R(\mathbf{A})$: space spanned by the columns of $\mathbf{A}$.

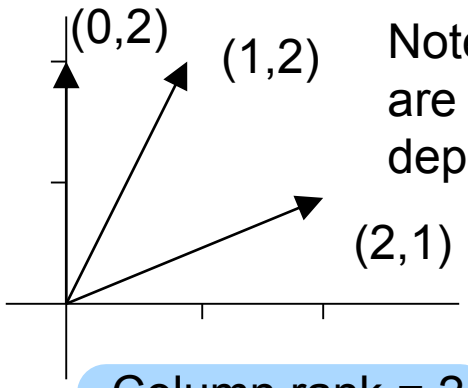
The row space $R(\mathbf{A}^t)$ is the space spanned by the rows of $\mathbf{A}$.

The row space and column space have the same dimensionality r, the *rank* of $\mathbf{A}$.

This fact sometimes expressed as "row rank = column rank". *

Example : $\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \end{bmatrix}$

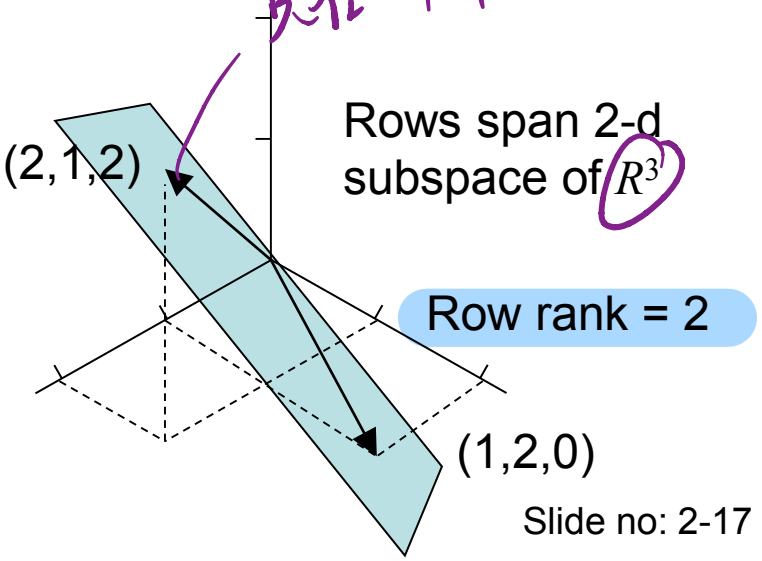
Columns span entire R^2



Note that columns are linearly dependent

Column rank = 2

只在一个平面内



Rows span 2-d subspace of R^3

Row rank = 2

Null Space; Orthogonal Subspaces

The *nullspace* $N(\mathbf{A})$ of $\mathbf{A}$ is the space *not* spanned by the rows of $\mathbf{A}$. This has dimension $n-r$.

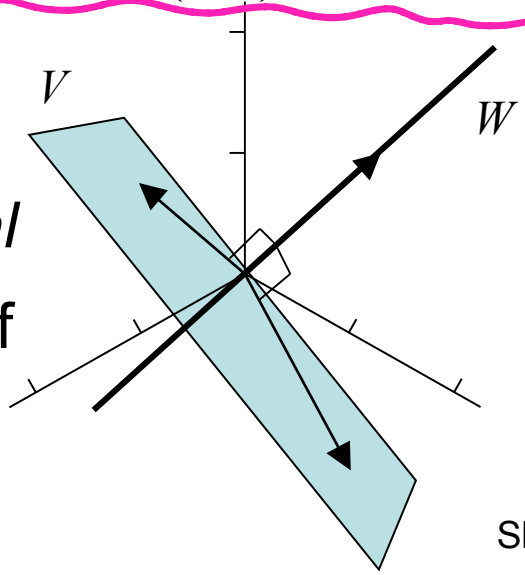
Two subspaces V and W are orthogonal if every vector $\mathbf{v}$ in V is orthogonal to every vector $\mathbf{w}$ in W.

I.e. we must have $\mathbf{v}^t\mathbf{w}=0$ for all $\mathbf{v}\in V$, $\mathbf{w}\in W$.

So, the nullspace $N(\mathbf{A})$ and row space $R(\mathbf{A}^t)$ are *orthogonal*.

Example: 2-d subspace V (plane)
is orthogonal to 1-d subspace W (line)

In the diagram, W is the *orthogonal complement* $V^\perp$ of V (the space of all vectors orthogonal to V).



Bases and Hilbert Spaces

Changing the nomenclature (but not the theory) to that of Transform Theory:

- A set of vectors or functions $\{\Psi_n\}$ spans a vector space if any element of that space can be expressed as a linear combination of members of that set

vector

$$s = \sum_n c_n \Psi_n$$

$\langle \Psi_n, \Psi_m \rangle = \delta_{nm}$

- $\{\Psi_n\}$ is a basis set if the c_n are unique.
- The set is an orthogonal basis if $n \neq m \Rightarrow \langle \Psi_n, \Psi_m \rangle = 0$
- The set is an orthonormal basis if $n = m \Rightarrow \langle \Psi_n, \Psi_m \rangle = 1$

That is, the Kronecker Delta Function

- A space which has all these properties is a type of Hilbert space
- Real numbers, complex numbers and integrable functions form Hilbert Spaces

Properties of Orthonormal Bases

- If $\{\psi_n\}$ constitutes a basis for V, then any vector or function in V can be written as

$$s = \sum_n c_n \Psi_n$$

- However, c_n may be difficult to compute. If $\{\psi_n\}$ form an orthonormal basis, this difficulty is eliminated, since then

$$c_n = \langle s, \Psi_n \rangle$$

- Thus if $\{\psi_n\}$ is a set of orthonormal basis for V, then any ***s*** in V can be written as

$$s = \sum_j \langle s, \Psi_j \rangle \Psi_j$$

$$= \langle s, \Psi_1 \rangle \Psi_1 + \langle s, \Psi_2 \rangle \Psi_2 + \ldots\ldots + \langle s, \Psi_n \rangle \Psi_n$$

Biorthogonal Bases

- Sometimes, orthonormal bases are not available. A generalization is the concept of **biorthogonal bases** , which are actually a pair of bases, each set not being orthogonal, but the two sets being orthogonal to each other (strictly speaking, they need to be normalised to satisfy the Kronecker Delta Function).

- If $\{\psi_n\}$ ("psi") and $\{\hat{\Psi}_n\}$ ("psi hat") are both basis vectors themselves for *V*, and satisfy

$$\langle \Psi_i, \hat{\Psi}_j \rangle = \delta_{ij}$$

then any **s** in *V* can be written as

$$s = \sum_{j=1}^{n} \langle s, \Psi_j \rangle \hat{\Psi}_j$$

Dual Base example

- **So, a basis set $\{\hat{\Psi}_i\}$ is said to be the dual basis of $\{\Psi_i\}$ if the** biorthogonality **condition**

$$\left\langle \Psi_i, \hat{\Psi}_j \right\rangle = \sum_k \Psi_i(k)\hat{\Psi}_j(k) = \delta_{ij}$$

$$= \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j \end{cases}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is satisfied. (The Kronecker Delta Function)

- **An example of a dual basis**

$$\{\Psi_i\} = \{(1,0),(1,1)\}$$

These vectors are not orthogonal.
They are linearly independent

$$\{\hat{\Psi}_i\} = \{(1,-1),(0,1)\}$$

These vectors are not orthogonal.
They are linearly independent

Show that these are biorthogonal:

Frames

- Sometimes, biorthogonal bases are not available either. We'd like to represent any vector in V as a linear combination of some simpler vectors / functions, while giving up orthogonality and even linear independence:

$$s = \sum_{j=1}^{n} \left\langle s, \Psi_j \right\rangle \hat{\Psi}_j = \sum_{j=1}^{n} \left\langle s, \hat{\Psi}_j \right\rangle \Psi_j$$

- Vectors $\{\psi_n\}$ form a **frame** with frame bounds *A*, *B* if for any vector *s* in V

$$A\|s\|^2 \le \sum_j \left| \left\langle s, \Psi_j \right\rangle \right|^2 \le B\|s\|^2$$

FS

Summary of the Fourier Series

- **Three forms**
  - **Original (sine and cosine components)**

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n.\cos(n.\omega.t) + \sum_{n=1}^{\infty} b_n.\sin(n.\omega.t)$$

  - **Cosine-with-phase form**

$$x(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \theta_k) \qquad -\infty < t < \infty$$

  - **Exponential form**

$$x(t) = \sum_{n=-\infty}^{\infty} X_n.\, e^{jn\omega t}$$

- **Dirichlet conditions**
- **Gibbs Phenomenon**

FT

Fourier Transform

- The Fourier Series can only be applied to periodic signals, giving an infinite number of discrete frequencies. However, periodic signals are non-informational.

- Non-periodic signals (signals containing information) cannot be analysed using the Fourier Series, the Fourier Transform (FT) is required.

- This gives us the bandwidth of a signal as the sum of an continuous infinity of sinusoids.

The Conditions for an FT

- **A signal is said to have a Fourier transform in the ordinary sense if the integral in the following equation converges (i.e. exists) .**

$$X(f) = \int_{t=-\infty}^{t=\infty} e^{-j\omega t} \cdot x(t)dt$$

x(t) is "well behaved" if:

1. the signal x(t) has a finite number of discontinuities, maxima, and minima within any finite interval of time.
2. if x(t) is absolutely integrable $\int_{t=-\infty}^{t=\infty} |x(t)|dt < \infty$

有限不连续 / 极值

条件 condition

These are the Dirichlet Conditions.

However some common signals are not absolutely integrable, such as a constant DC signal. But in practice this could not exist for an infinite time.

generally not suitable for use as basis functions

Inverse Fourier Transform

Given a signal x(t) with Fourier transform X(f), x(t) can be recomputed from X(f) by application of the inverse Fourier transform give by:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(f)\cdot e^{j\omega t}\,df$$

To denote the fact that X(f) is the Fourier transform of x(t), or that X(f) is the inverse Fourier transform of x(t), the transform pair notation:

$$x(t) \leftrightarrow X(f)$$

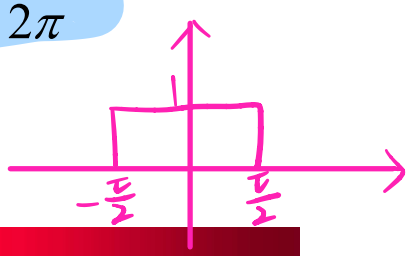
will sometimes be used.

$\mathrm{sinc}(aw) = \frac{\sin(a\pi w)}{a\pi w}$

One of most fundamental transform pairs in the Fourier Theory is the pair

$$p_\tau(t) \leftrightarrow \tau\,\mathrm{sinc}\frac{\tau\omega}{2\pi}$$

矩形波



Fourier Transform: Hermitian Symmetry

Expand the Fourier transform of a function $s(t)$ into the sum of a odd and an even function:

$$S(\omega) = \int_{-\infty}^{\infty} [s_e(t) + s_o(t)]\ [\cos(\omega t) - j\sin(\omega t)]\, dt$$

$$= \int_{-\infty}^{\infty} s_e(t)\cos(\omega t)dt + \int_{-\infty}^{\infty} s_o(t)\cos(\omega t)dt - j\int_{-\infty}^{\infty} s_e(t)\sin(\omega t)dt - j\int_{-\infty}^{\infty} s_o(t)\sin(\omega t)dt$$

$$= \int_{-\infty}^{\infty} s_e(t)\cos(\omega t)dt - j\int_{-\infty}^{\infty} s_o(t)\sin(\omega t)dt$$

If even, Fourier transform is real $S_e(\omega) = \mathrm{Re}[S_e(\omega)]$

If even, Fourier transform is even $S_e(\omega) = S_e(-\omega)$

If odd, Fourier transform is imaginary $S_o(\omega) = \mathrm{Im}[S_o(\omega)]$

If odd, Fourier transform is odd $S_o(\omega) = -S_o(-\omega)$

Hermitian or Conjugate Symmetry:

$$S(\omega) = S_e(\omega) + S_o(\omega) = S_e(-\omega) - S_o(-\omega)$$

$$= [S_e(-\omega) + S_o(-\omega)]^* = S^*(-\omega)$$

A hermitian function is a complex function with the property that its complex conjugate is equal to the original with the variable changed in sign.

Linearity

If $s_1(t)$ has Fourier transform $S_1(w)$ and $s_2(t)$ has Fourier transform $S_2(w)$, then

$a \cdot s_1(t) + b \cdot s_2(t)$ has Fourier transform

$$\int_{-\infty}^{\infty} [a \cdot s_1(t) + b \cdot s_2(t)] e^{-j\omega t} dt$$

$$= a \int_{-\infty}^{\infty} s_1(t) e^{-j\omega t} dt + b \int_{-\infty}^{\infty} s_2(t) e^{-j\omega t} dt$$

$$= a \cdot S_1(\omega) + b \cdot S_2(\omega)$$

Similarly for the inverse Fourier transform

Scaling

If $s(t)$ has Fourier transform $S(w)$ then $s(a \cdot t)$ has Fourier transform $\frac{S(\omega / a)}{|a|}$

an inverse scaling of the frequencies proportional to the scaling in time

square ✗

$a > 0 \rightarrow$

$$\int_{-\infty}^{\infty} s(a \cdot t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} s(x) e^{-j\omega x / a} d(x / a)$$

$$= \frac{1}{a} \int_{-\infty}^{\infty} s(x) e^{-j\omega x / a} dx = \frac{S(\omega / a)}{a}$$

$a < 0 \rightarrow$

$$\int_{-\infty}^{\infty} s(a \cdot t) e^{-j\omega t} dt$$

$$= \int_{\infty}^{-\infty} s(x) e^{-j\omega x / a} d(x / a)$$

$$= \frac{-1}{a} \int_{-\infty}^{\infty} s(x) e^{-j\omega x / a} dx = \frac{-S(\omega / a)}{a}$$

Also known as the Scaling Theorem

The Shift Property

If $s(t)$ has Fourier transform $S(w)$ then

$s(t)e^{j\omega_0 t}$ has Fourier transform

$$\int_{-\infty}^{\infty} s(t)e^{j\omega_0 t}e^{-j\omega t}dt$$

$$= \int_{-\infty}^{\infty} s(t)e^{-j(\omega-\omega_0)t}dt$$

$$= S(\omega-\omega_0)$$

Thus multiplication of $s(t)$ by $e^{j\omega_0 t}$ shifts the frequency spectrum of $s(t)$ so that it is centred on the point $\omega = \omega_0$ in the frequency domain (that is, MODULATION of a carrier by a cisoid).

The Delay Property

If $s(t)$ has Fourier transform $S(w)$ then the delayed version $s(t-a)$, has Fourier transform:

$$\int_{-\infty}^{\infty} s(t-a)e^{-j\omega t}dt = \int_{-\infty}^{\infty} s(t)e^{-j\omega(t+a)}dt$$

$$= e^{-j\omega a}\int_{-\infty}^{\infty} s(t)e^{-j\omega t}dt = e^{-j\omega a}S(\omega)$$

This implies that delaying a signal by a time a causes its Fourier transform to be multiplied by $e^{-j\omega a}$.

Symmetry

If $s(t)$ has Fourier transform $S(\omega)$ then $S(t)$ has Fourier transform $2\pi s(-\omega)$

$$s(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\, e^{j\omega t}\, d\omega \qquad \text{Replace } \omega \text{ with } y$$

$$2\pi s(t) = \int_{-\infty}^{\infty} S(y)\, e^{jyt}\, dy \qquad \text{Replace } t \text{ with } -\omega$$

$$2\pi s(-\omega) = \int_{-\infty}^{\infty} S(y)\, e^{-jy\omega}\, dy \qquad \text{Replace } y \text{ with } t$$

$$2\pi s(-\omega) = \int_{-\infty}^{\infty} S(t)\, e^{-jt\omega}\, dt$$

So if $s(t)$ and $S(w)$ form a Fourier transform pair then $S(t)$ and $2\pi s(-\omega)$ form another Fourier transform pair.

Convolution Theorem

$$f(u) * g(u) \equiv (f * g)(u) \equiv \int_{-\infty}^{\infty} f(x)g(u-x)dx$$

If $s_1(t)$ has Fourier transform $S_1(w)$ and $s_2(t)$ has Fourier transform $S_2(w)$, then the Fourier transform of the convolution $s_1(t)*s_2(t)$ is given by:

$$\int_{-\infty}^{\infty} [s_1(t) * s_2(t)] e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-j\omega t} \left[\int_{-\infty}^{\infty} s_1(\tau) s_2(t-\tau) d\tau \right] dt$$

$$= \int_{-\infty}^{\infty} s_1(\tau) \left[\int_{-\infty}^{\infty} s_2(t-\tau) e^{-j\omega t} dt \right] d\tau = \int_{-\infty}^{\infty} s_1(\tau) \left[\int_{-\infty}^{\infty} s_2(x) e^{-j\omega(x+\tau)} dx \right] d\tau$$

$$= \int_{-\infty}^{\infty} s_1(\tau) e^{-j\omega\tau} d\tau \int_{-\infty}^{\infty} s_2(x) e^{-j\omega x} dx = S_1(\omega) S_2(\omega)$$

That is, the Fourier transform of the convolution is the product of the Fourier transforms.

Sampling

Shannon Sampling Theorem

- Continuous-time signal $x(t)$ with frequencies no higher than f_{max} can be reconstructed from its samples $x(k\,T_s)$ if samples taken at rate $f_s > 2\,f_{max}$

  Nyquist rate = $2\,f_{max}$

  Nyquist frequency = $f_s / 2$

Critical Sampling if

$$f_s = 2\,f_{max}$$

- Example: Sampling audio signals

  Human hearing is from about 20 Hz to 20 kHz

  Apply lowpass filter before sampling to pass frequencies up to 20 kHz and reject high frequencies

  Lowpass filter needs 10% of maximum passband frequency to roll off to a sufficiently small value (2 kHz rolloff in this case), hence high order filter.

Aliasing

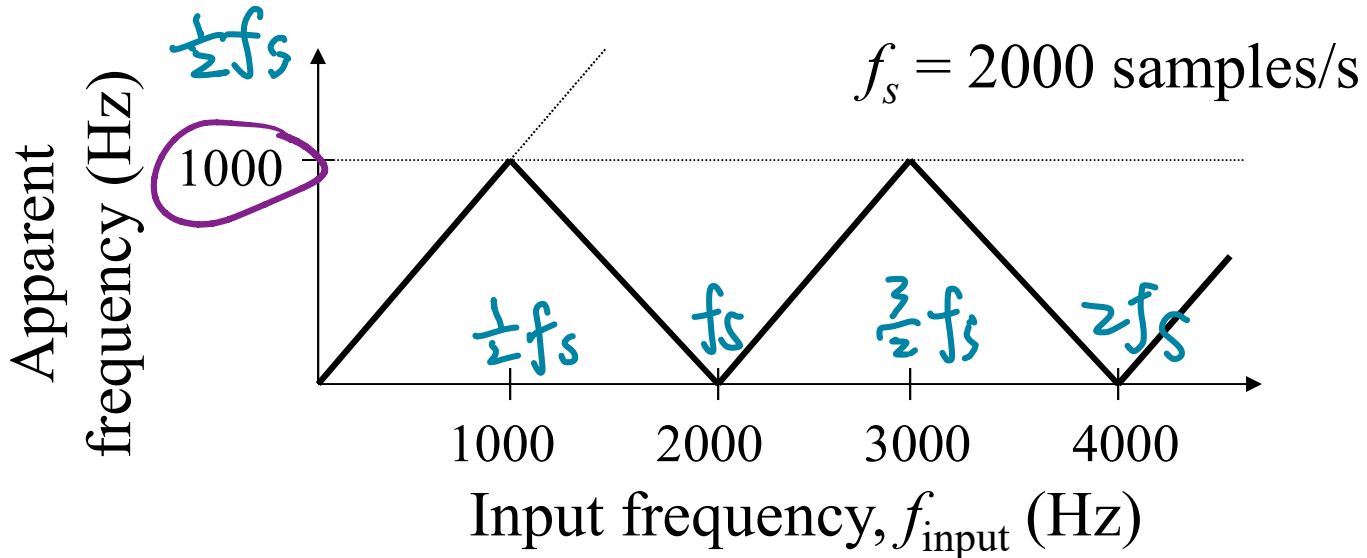
- Since l is any integer, a countable but infinite number of sinusoids will give same sequence of samples
- Frequencies $f_0 + l f_s$ for $l \neq 0$ are called aliases of frequency f_0 with respect to f_s

  All aliased frequencies appear to be the same as f_0 when sampled by f_s

can make high frequency noise appear part of the signal's bandwidth

Aliasing and Folding

- Aliasing and folding of a sinusoid $\sin(2\pi f_{input} t)$ sampled at $f_s = 2000$ samples/s with f_{input} varied



- Mirror image effect about $f_{input} = \frac{1}{2} f_s$ gives rise to name of folding

So if sampled at 2000 samples/s, 1000Hz appears as 1000Hz; 2000Hz appears as 0Hz; 3000Hz as 1000Hz; 3500Hz as 500Hz; etc.

Sampling a signal in the time domain results in multiples of the baseband frequency spectrum
separated by integer multiples of the sampling frequency.
Briefly explain how aliasing can be prevented.

1. A bandlimiting filter must be used prior to sampling [1 mark]. This is a lowpass filter [1 mark]
2. Sampling frequency must be greater than twice the bandwidth of the signal to give a gap between spectra [1 mark].
3. The filter cutoff frequency should be set at the upper limit of the baseband spectrum [1 mark].
4. The roll-off rate of the filter must be sufficiently great to provide adequate attenuation at the lower end on the first replicated spectrum [1 mark].

Explain what effect in the frequency domain of sampling a signal in the time domain.

Sampling in the time domain is multiplication of the signal with a comb of Dirac delta functions [1 mark]. This is equivalent to convolution with a comb of delta functions in the frequency domain. [1 mark].
The result of this is that sampling replicates the frequency spectrum of the continuous time signal in the frequency domain at inter multiple of the sampling frequency [1 mark].
There is a gap between spectra if the signal is oversampled [1 mark], spectra are adjacent if the signal is critically sampled (Nyquist sampling rate) [1 mark], spectra overlap if the signal underdamped[1 mark].
Aliasing occurs when the signal is sampled at less than the Nyquist limit [1 mark]. Aliasing is when an unwanted signal is mistaken for a wanted one [1 mark]. Once aliasing occurs it is not possible to remove it [1 mark].
The baseband spectrum can be obtained from the bandpass spectrum by using a low-pass filter [1 mark].

DFT

Discrete-Time Fourier Transform

- Forward transform of discrete-time signal $x[n]$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

  - Assumes that $x[n]$ is two-sided and infinite in duration
  - Produces $X(\omega)$ that is continuous and periodic in ω (in units of rad/sample) with period 2π due to exponential term

So the DTFT is not practical

- Inverse discrete-time Fourier transform

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

Discrete Fourier Transform

- Discrete Fourier transform (DFT) of a discrete-time signal $x[n]$ with finite extent $n \in [0, N\text{-}1]$

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}nk} = X(\omega)\Big|_{\omega=\frac{2\pi}{N}k} \quad \text{for } k = 0, 1, \ldots, N\text{-}1$$

  - $X[k]$ is periodic with period N due to exponential
  - DFT assumes $x[n]$ is also periodic with period N

- Inverse discrete Fourier transform

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\, e^{j\frac{2\pi}{N}nk}$$

- "Twiddle factor"

$$W_N = e^{j\frac{2\pi}{N}} \Rightarrow x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k] W_N^{nk}$$

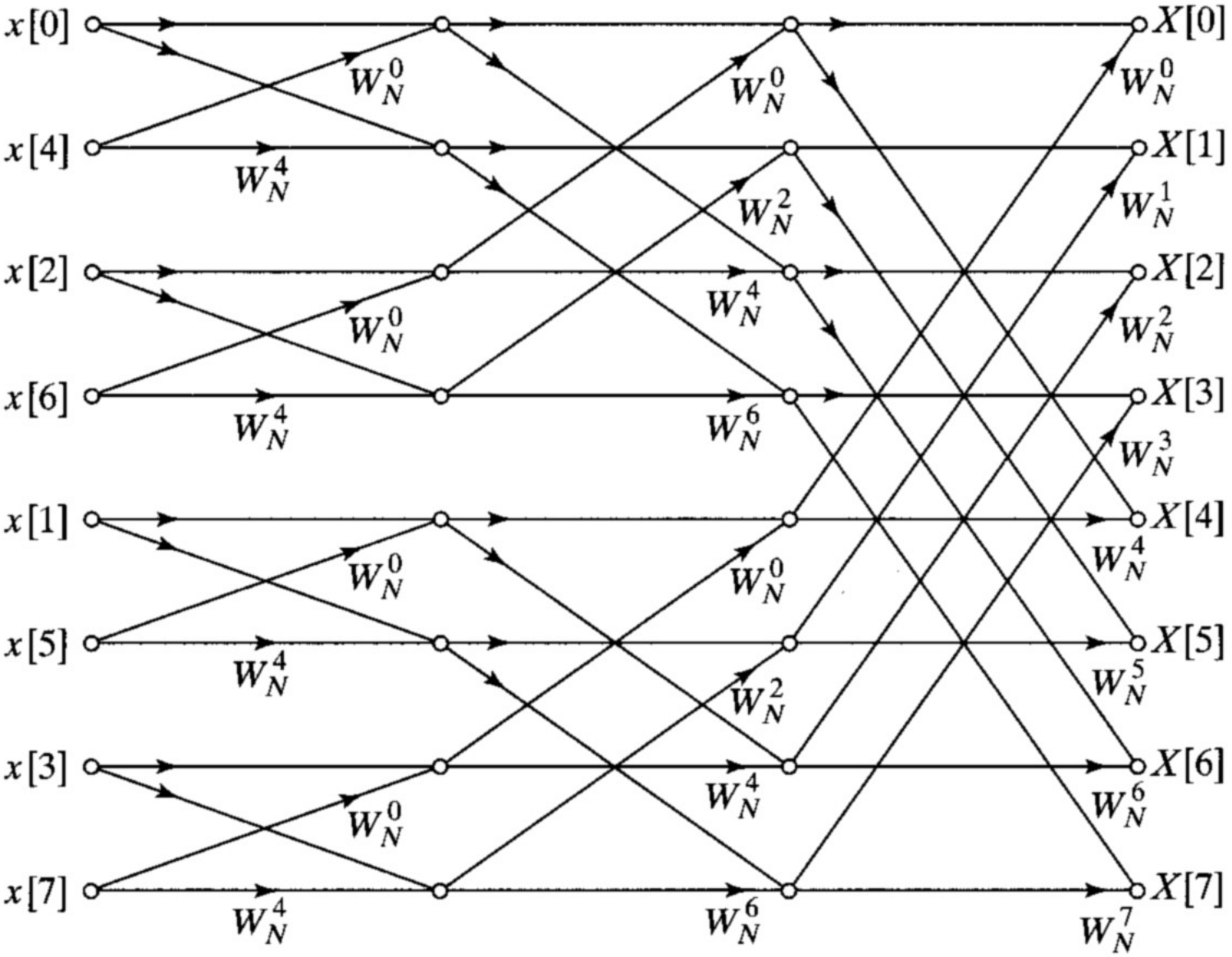
N^2 complex multiplications
$N\,(N-1)$ complex additions
N^2 integer multiplications
N^2 modulo indexes into lookup table of twiddle factors

DFT in Practice

- The DFT is the only practical way to find the FT of a signal (any other version of the FT requires an infinite amount of calculation).
- The DFT assumes that the sequence to be transformed is periodic.
- Suppose we want to find the DFT of a piece of music.
- If we sample the whole piece of music we will find its accurate DFT.
- If we take a segment of the music, we can find the segment's DFT.
- But the DFT assumes that the sequence is periodic and this is very unlikely to be the case for the music.
- So if we find the DFTs for each segment of the music, the result will not be an accurate FT of the music.
- Finding the DFT of the whole music is accurate but will take a long time because there are many samples.
- The shorter each segment (fewer samples) the faster the calculation but the less accurate the transform.

the longer sequence length
the better the resolution

The complete 8-point decimation-in-time FFT



Now let's take a closer look at the 2-point DFT

- The expression for the 2-point DFT is:

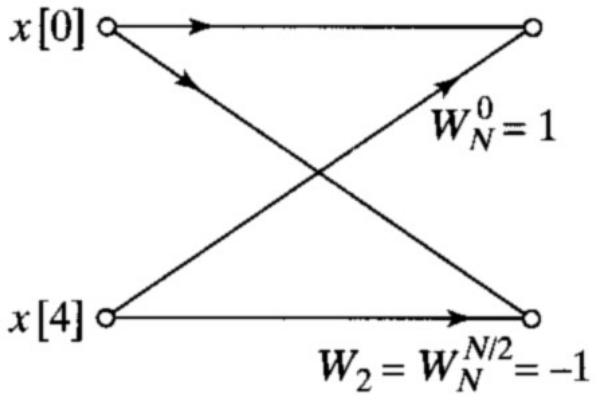
$$X[k] = \sum_{n=0}^{1} x[n] W_2^{nk} = \sum_{n=0}^{1} x[n] e^{-j2\pi nk/2}$$

- Evaluating for $k = 0,1$ we obtain

$$X[0] = x[0] + x[1]$$

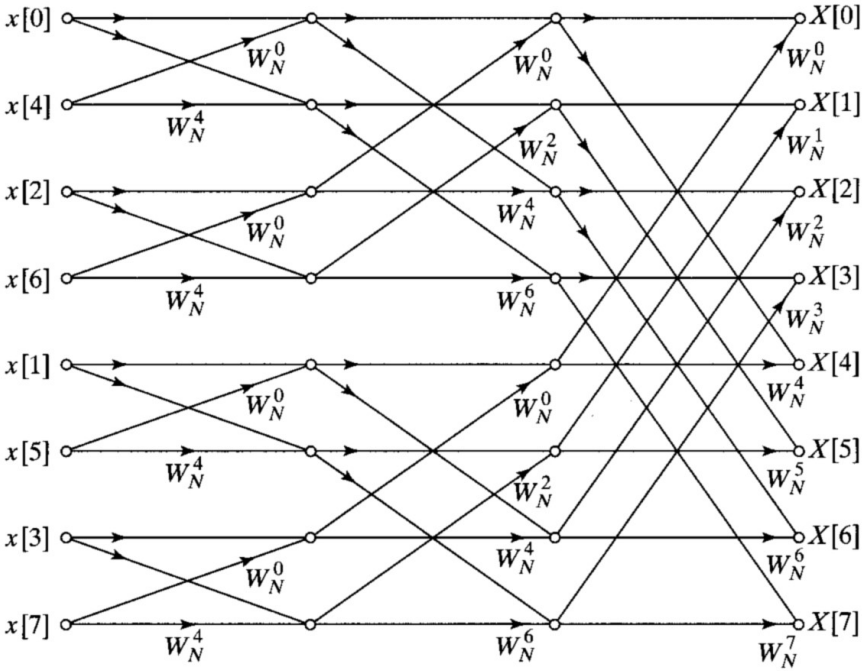
$$X[1] = x[0] + e^{-j2\pi 1/2} x[1] = x[0] - x[1]$$

which in signal flowgraph notation looks like ...



This topology is called the basic "butterfly"

Number of multiplies for N-point FFTs



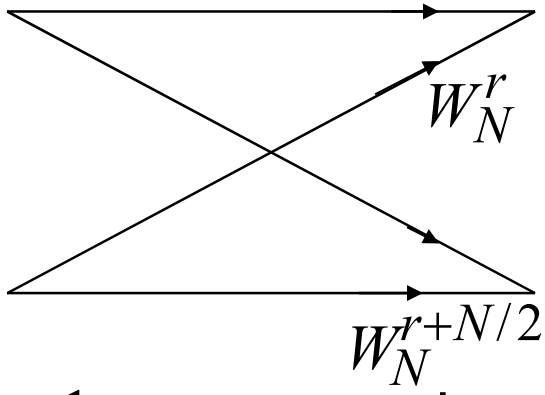
- Let $N = 2^{\nu}$ where $\nu = \log_2(N)$
- ($\log_2(N)$ columns)($N/2$ butterflys/column)(2 mults/butterfly)

  or ~ $N \log_2(N)$ multiplies

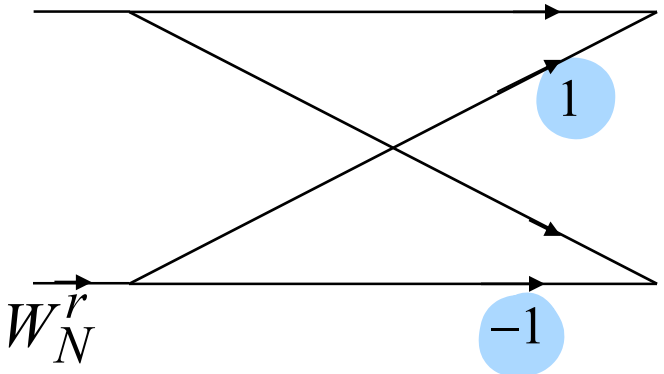
∵ FFT "decimate" the input sequence in powers of 2.

Additional timesavers: reducing multiplications in the basic butterfly

- As we derived it, the basic butterfly is of the form

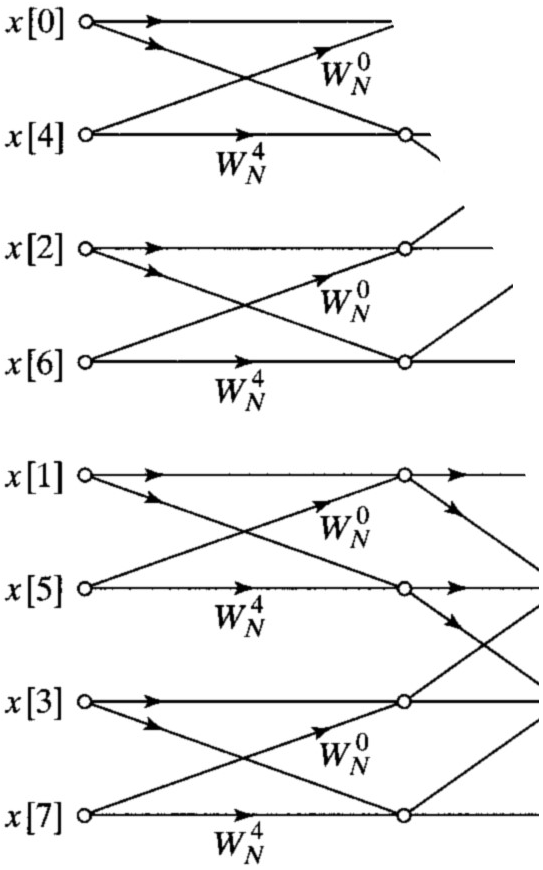


- Since $W_N^{N/2} = -1$ we can reduce computation by 2 by premultiplying by W_N^r



Bit reversal of the input

- Recall the first stages of the 8-point FFT:



Consider the binary representation of the indices of the input:

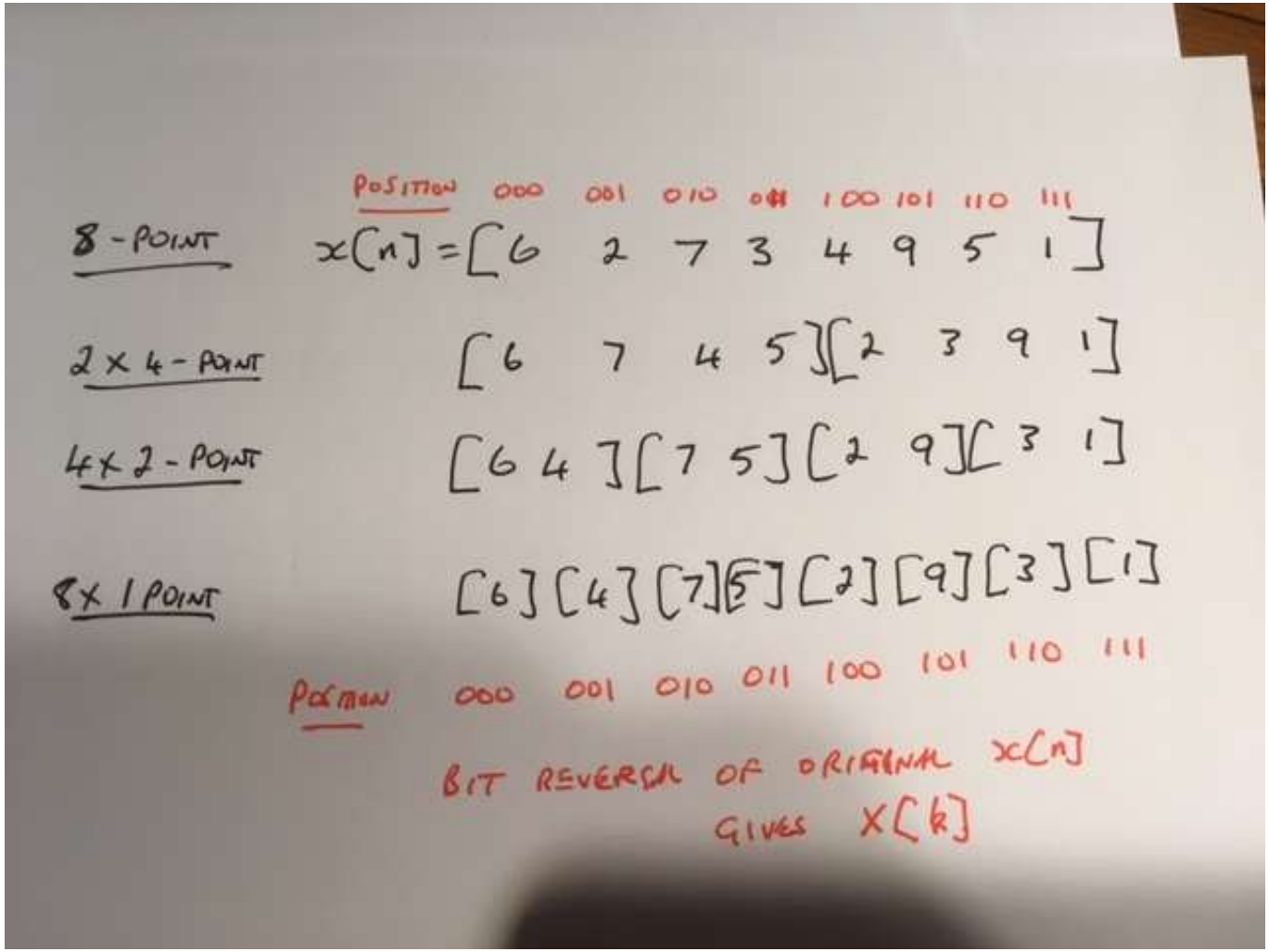
0	000
4	100
2	010
6	110
1	001
5	101
3	011
7	111

If these binary indices are time reversed, we get the binary sequence representing

0,1,2,3,4,5,6,7

Hence the indices of the FFT inputs are said to be in bit-reversed order

Example with an 8-point input sequence



Describe the process of implementing an FFT. Refer to radix-2 decimation in time. State how the reverse process is carried out.
[3, -4, 8, -2, -7, 9, 13, 6]

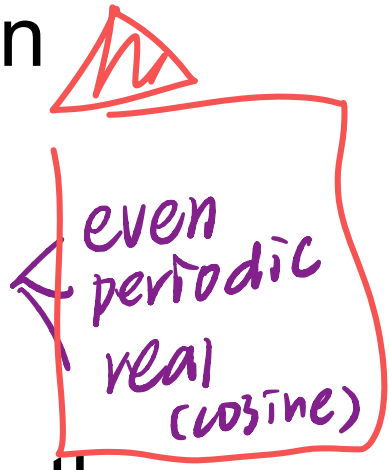
A radix-2 sequence is one whose number of elements is a power of 2 [1 mark]. A sequence is split into two [1 mark], one of which is the even numbered elements and one the odd numbered elements [1 mark]. This process is continued till we have individual elements [1 mark].
For example, for the 8-point signal as follows:
[3, -4, 8, -2, -7, 9, 13, 6]
This is split into:
[3 8 -7 13] and [-4 -2 9 6]
And so on:
[3 -7] [8 13] [-4 9] [-2 6]
[3] [-7] [8] [13] [-4] [9] [-2] [6] [3 marks, 1 for each row]
Re-ordering the sequence in this way can be performed using bit-reversal, as each position is the reverse of the binary value of the original position. Element values are swapped accordingly, e.g., the value in position 3 (binary 011) [-2] is swapped with the value in position 6 (binary 110) [13] [1 mark].
Each 1-point signal is then transformed to the frequency domain, nothing is required to do this step [1 mark].
The individual frequency values are then combined by reversing, in steps, the decomposition process: [1 mark]
In the above case, the 16 individual frequency points are grouped into 8 groups of 2 points each. This is carried out by shifting, adding zeroes in the gaps and then adding. [1 mark]

DCT

Discrete Cosine Transform DCT

As with the DFT, we assume that the input sequence is periodic in order to obtain an accurate Fourier Transform.

In addition, for the DCT the sequence is assumed to be even and periodic.



We will consider the DCT Type II. This is the most common, and is used as a 2D transform to produce JPEG compressed images. (There are 8 versions altogether)

Discrete Cosine Transform DCT

Summary: advantages:

1. • Reversible (lossless) transform that represents a discrete signal as a set of cosine coefficients (DCT basis functions are orthogonal).
2. • Similar to DFT, but uses only cosines and therefore avoids any complex numbers.
3. • Real Input, real output
   • We will consider only discrete with number of input= number of output (n=k).

4. implementation is fast because only real calculations.
5. most energy concentrated in a few coefficients, so can discard some coefficients while keeping most of signal.

1-Dimensional DCT

$$DCT[k] = c(k)\sum_{n=0}^{N-1} s[n]\cos\frac{\pi(2n+1)k}{2N} \qquad k = 0,1,2\ldots N-1$$

$$DCT[k] = \langle s, \psi_k \rangle$$

$$c(k) = \begin{cases} \sqrt{1/N} & k = 0 \\ \sqrt{2/N} & k \neq 0 \end{cases}$$

c(k) is the normalisation factor.

- Orthonormal

$$\langle \psi_m, \psi_n \rangle = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

The Basis Functions Ψ_k are the cosine terms in the definition.
They are calculated for each value of k, with n = 0….N-1

8x8 DCT Transform Matrix

The coefficients in each row of the transform matrix are the amplitudes of 8 samples of a cosine wave. The first row is a cosine wave of 0Hz (DC), then the frequencies of each cosine wave are increasing (AC).

$$\Psi = \frac{1}{2}\begin{bmatrix} .71 & .71 & .71 & .71 & .71 & .71 & .71 & .71 \\ .98 & .83 & .56 & .19 & -.19 & -.56 & -.83 & -.98 \\ .92 & .38 & -.38 & -.92 & -.92 & -.38 & .38 & .92 \\ .83 & -.19 & -.98 & -.56 & .56 & .98 & .19 & -.83 \\ .71 & -.71 & -.71 & .71 & .71 & -.71 & -.71 & .71 \\ .56 & -.98 & .19 & .83 & -.83 & -.19 & .98 & -.56 \\ .38 & -.92 & .92 & -.38 & -.38 & .92 & -.92 & .38 \\ .19 & -.56 & .83 & -.98 & .98 & -.83 & .56 & -.19 \end{bmatrix}$$

2D & nD Transforms

- Many transforms are applied in 1 Dimension.
- But some can also be applied in 2D or *n*D. E.g. Fourier:

1-D: $S(\omega) = \int_{-\infty}^{\infty} s(t) e^{-j\omega t} dt$

2-D: $S(w,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} s(x,y) e^{-j(wx+vy)} dxdy$

$$= \int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} s(x,y) e^{-jwx} dx\right) e^{-jvy} dy$$

n-D: $S(\mathbf{w}) = \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} s(\mathbf{x}) e^{-j(\mathbf{w}\cdot\mathbf{x})} d\mathbf{x}$

Dot Product

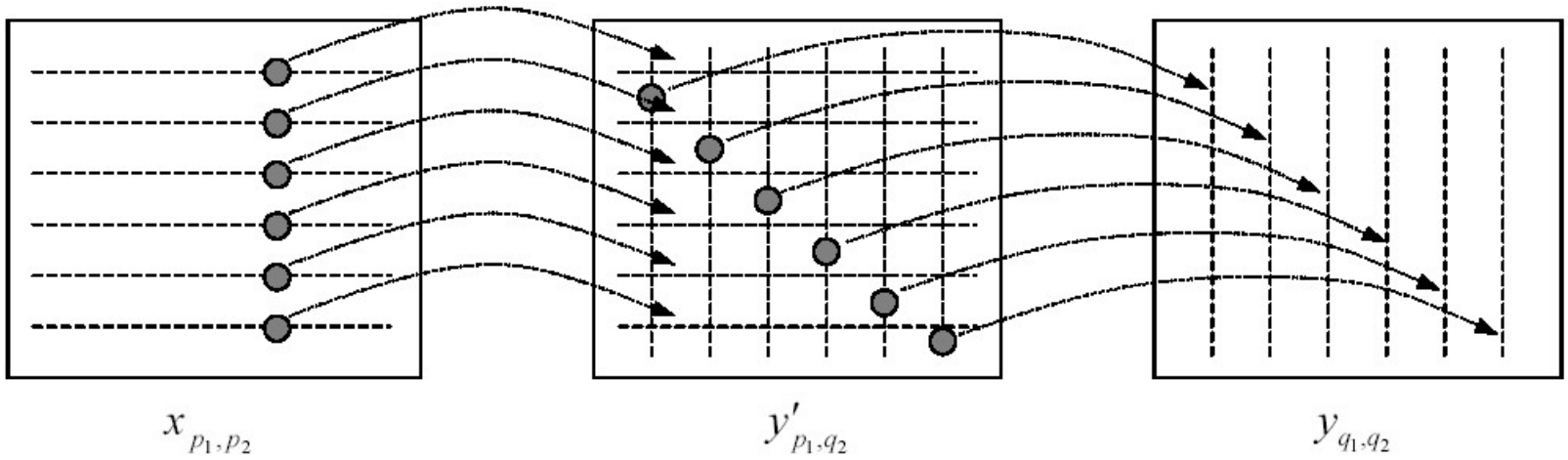
where $\mathbf{w} = (w_1, w_2, \ldots, w_n)$ and $\mathbf{x} = (x_1, x_2, \ldots, x_n)$.

An n-D transform is *separable* if we can apply sequence of 1-D transforms (see 2-D case above).

Separable DCT applied to Image Compression

Separable Transforms

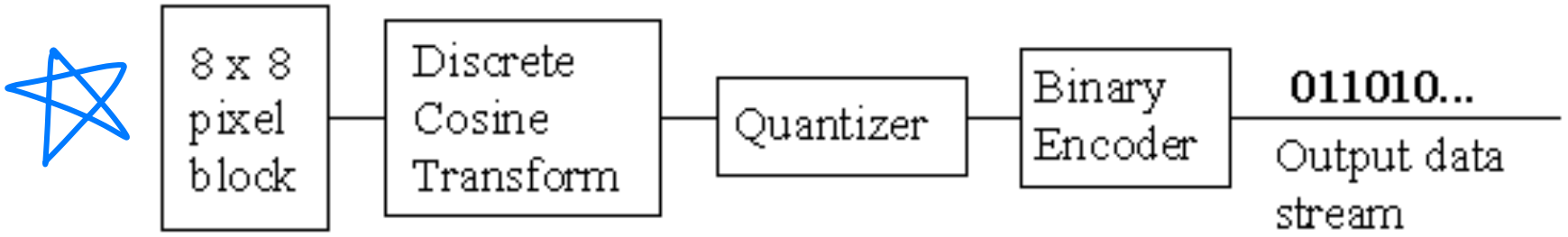
May be implemented by applying the one dimensional transform first to the rows of the image and then to its columns (note that changing the application order does not change the result).



Each row of the 2D image is transformed individually as a 1D transform

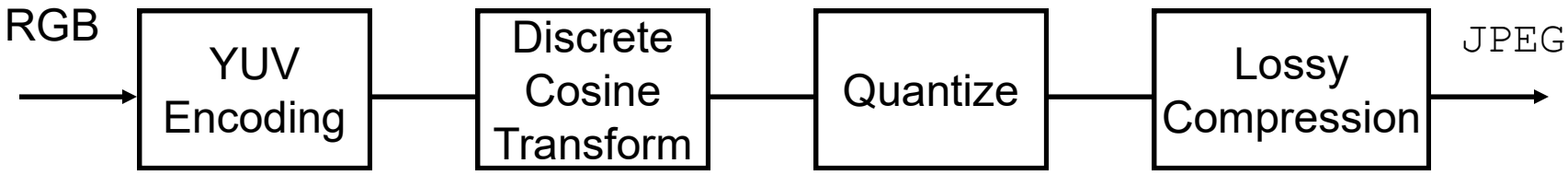
Then the columns of this 1D transform are transformed to produce the 2D transform data

JPEG Compression - Macroblocks



- YUV encoded image (like RGB)
- Y is cut into 8x8 tiled pixel regions.
- U and V cut into 8x8 tiled pixel regions.
- Macroblock defined as 4 Y tiles that form a 16x16 pixel region and associated U and V tiles.
- Macroblocks organized in row order fashion from top to bottom.

JPEG Encoding Steps



Encoding

- **Convert to different colour representation**
- **Typically get 2:1 compression**

Discrete Cosine Transform (DCT) (Lossless)

- **Transform 8 X 8 pixel blocks** separately

Quantize

- **Reduce precision of DCT coefficients**
- **Lossy step**

Lossy Compression

- **Express image in highly compressed form**

What's lossless compression and lossy compression?

i) Lossless compression is where fewer bits are required to store the data but no information is discarded. [1 mark]. This is achieved by eliminating redundancy [l mark]
ii) Lossy compression results in a loss of information [1 mark]. This is achieved by discarding marginally important information [1 mark].

Explain in what context the Discrete Cosine Transform (DCT) is suitable for lossy data compression.

In the DCT, most of the signal information tends to be concentrated in a few low frequency components. [1 mark] The amount of information that can be allowed to be lost during compression depends on the application. [1 mark] In visual images, much of the high frequency information can be discarded because the human eye is far less sensitive to them than the lower frequency information. [1 mark] The DCT itself is not lossy [1 mark]. But it does separate the frequency components to allow compression to take place.

Why is the DCT more efficient than the Discrete Fourier Transform (DFT).

Because the DCT is based on the cosine function, it is the real part of the DFT and complex arithmetic is not required in carrying out the DCT [1 mark]. The FT of a real and even function is real and even, meaning the DCT is efficient for real data with even symmetry [1 mark].

The Discrete Cosine Transform (DCT) transforms functions of time (discrete signals) and functions of position (images) into functions of frequency. Explain how it does this.

The DCT correlates the input data [1 mark] with a series of cosine functions of increasing frequency, starting with dc [1 mark]. These discrete cosine functions are the basis functions. Correlation gives a large value of output when there is a similarity between the input data distribution and a particular basis function [1 mark]. Therefore, the output sequence from the DCT is related to the frequency of the distribution of data in the input sequence [1 mark].

TFA

Time-Frequency Analysis

This example shows some of the applications of time-frequency analysis, including:

- Feature extraction (e.g. the short bursts of signal)….looking for particular events and at what times they happen, e.g. electrocardiograms (ecgs), hydrocarbon exploration, etc.

and information from the signal

- Trend analysis (e.g. the change of information with time), e.g. financial information, environmental data (weather, etc), etc.

Ignore the short features
look at what the signal is doing over a long term

Fourier Transform Limitation

- We see that the Fourier Transforms of these two different signals are identical.
- The Fourier Transforms tell us the range of frequencies contained in the signals.
- But if we wanted to know how the frequencies changed with time then the Fourier Transform could not give that information.
- Time-frequency analysis studies both the time and frequency domains simultaneously.

Limitation of Fourier Transform

We know that:

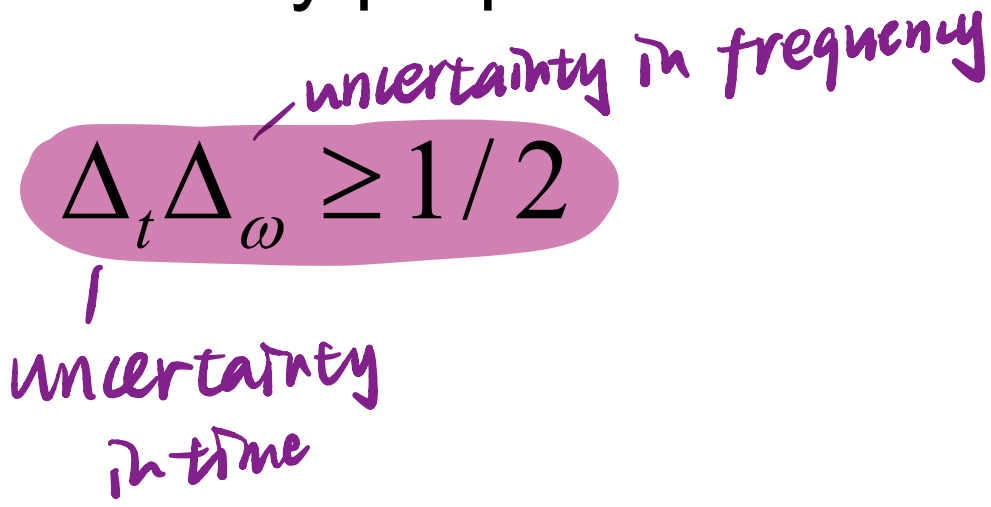
- The basis functions (complex sinusoids) are spread over the entire time domain.
- Loses all time information since integration is performed over all times.
- Not possible to discriminate signals that have the same frequency but occurring at different times.
- Signals suitable for Fourier transform are time invariant (stationary).
- Applications suitable for the Fourier transform are those which concern frequency only.

So what can we about it for non-stationary signals? → apply a window. → localise frequency in time

for the non-stationary signal, FT can provide details of the frequencies present in a signal, but it doesn't tell when those frequencies occurred.

Uncertainty Principle

- The Uncertainty Principle states:
- The product of the uncertainty in time and the uncertainty in frequency is a constant.
- So the are inversely proportional.



So far we have

$$\Delta_t^2 \Delta_\omega^2 \geq \left(-\frac{1}{2} \int_{-\infty}^{\infty} s^2(t) dt \right)^2$$

But $\int_{-\infty}^{\infty} s^2(t) dt = 1$ since it is the energy of the signal,

so $\Delta_t^2 \Delta_\omega^2 \geq \left(-\frac{1}{2} \times 1 \right)^2$ i.e. $\underline{\underline{\Delta_t \Delta_\omega \geq \frac{1}{2}}}$

Which was what we wanted.

Finally, Schwartz' inequality is exact (an equality) when the two functions ψ_1 and ψ_2 are colinear, $\psi_1 = c\psi_2$

With our functions $\psi_1 = ts(t)$ and $\psi_2 = \frac{d}{dt} s(t)$

this happens for Gaussian $s(t) = Ae^{-\alpha t^2}$,

$$\psi_2 = \frac{d}{dt} s(t) = -2\alpha t A e^{-\alpha t^2} = -2\alpha t s(t) = -2\alpha \psi_1$$

So $\Delta_t \Delta_\omega = \frac{1}{2}$ for a Gaussian

bell-shaped

a c gives the best resolution for any STFT

Δt is small, then Δw is large
Δw is small, then Δt is large

Uncertainty Principle

- We can easily find signals normalised in both time and frequency simultaneously, but it is not possible to *localise* a function in time and frequency simultaneously. (accurately at the same time)
- Different definitions of the Fourier Transform yield different versions of the Uncertainty Principle.
- The Time-Bandwidth Product $\Delta_t \Delta_\omega$ is a measure of the pulse complexity 脉冲复杂度
- A signal fixed in time has infinite bandwidth
- A signal fixed in frequency has infinite duration
- Discovered by Heisenberg and Applied to quantum mechanics

Briefly explain the uncertainty principle an discuss its implications in the design of transforms for time-frequency analysis.

b) Simply, the Heisenberg Uncertainty Principle in the context of time-frequency analysis states that it is not possible to know accurately both the time location and frequency value at that time [1 mark]. There is an uncertainty in both time and frequency, and the product of the uncertainties is a constant [1 mark]. If Δ_t is the uncertainty in time and Δ_ω is the uncertainty in frequency, it can be shown that (although the value of the constant depends on the definition of the Fourier Transform that is used in its proof):

$\Delta_t \Delta_\omega \geq 1/2$ [1 mark]

In time-frequency analysis, we use a moving window to investigate the frequency content of a signal at different time locations [1 mark]. The longer the duration of the window the more accurately the frequency content is known but the less accurately the time location is known [1 mark]. The shorter the duration of the window the more accurately time location is known but the less accurately the frequency [1 mark]. The shape of the window function affects the accuracy of the results [1 mark]. We need a function that dies away quickly and also has a limited frequency content [1 mark]. It can be shown that the only function minimising the product of Δ_t and Δ_ω is a Gaussian function

$g(t) = e^{-at^2}$ [1 mark]

The Gabor STFT uses a Gaussian function as its window [1 mark].

STFT Features of STFT

- Unlike FT, the STFT will give the information of frequency at different time intervals.
- Uncertainty Principle holds- The product of time resolution and frequency resolution is always greater than a minimum value.

$$\Delta_t \Delta_\omega \geq 1/2$$

- It is not possible to know exactly what frequency components exist at a particular point in time.
- What we can know are the time intervals in which certain bands of frequencies exist.
- Narrow window means good time resolution but poor frequency resolution and wide window means poor time resolution but good frequency resolution.
- For a given window, time resolution is fixed.

the problem is fixed by the product of uncertainty in t/f is a constant

width

若要改变分辨率，则需要重新选择窗

width narrow enough to let signal within window be stationary. but poor localisation in f domain. Tradeoff. The Uncertainty Principle limit the t/f localisation

Window Transform

- We can see that if the window is very wide, then we cannot localise the frequency content very accurately in time.
- However we will be able to know the frequencies present during the window quite accurately because we will have many cycles of them to be able to calculate the frequencies.

Window Transform

- If the window is very narrow then the localisation in time is more accurate
- However the knowledge of the frequencies present will not be accurate because we will "see" only part of the cycle of the waveform and will not be able to know the frequency accurately.
- If I look out my window and see the road outside I cannot know how long the road is because the window is too narrow.

The definition of WFT: is where a window is translated across the signal being transformed so that frequency information can be localised in time. The FT is evaluated for the portion of signal within the window.

The Windowed Fourier Transform

- **Harmonic wave e^{-jwt}**

(to perform the FT)

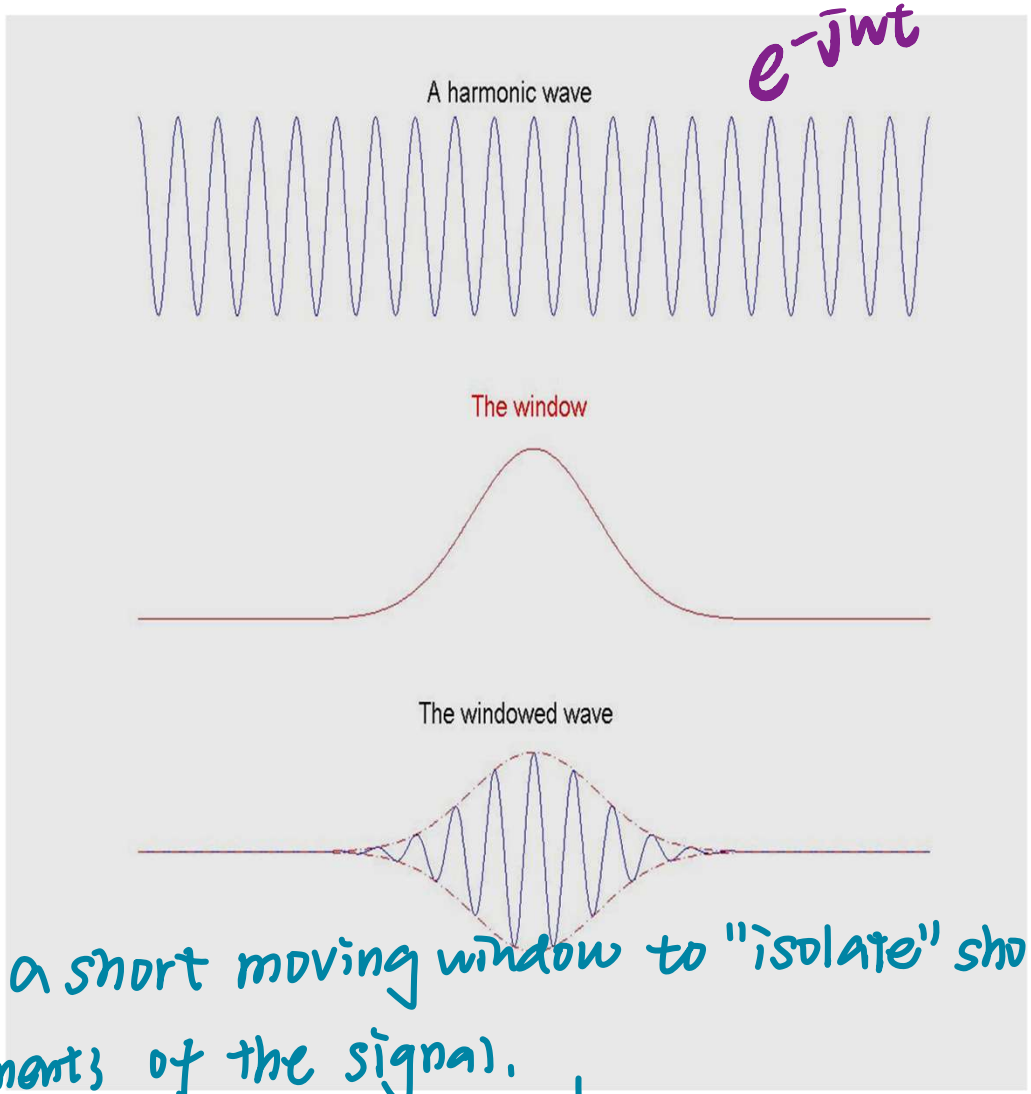
- **A window $\gamma(t)$**

(this will be moved across the signal)

- **A windowed wave**

$\gamma(\tau-t)\ e^{-jwt}$

(the basis function)



e^{-jwt}

use a short moving window to "isolate" short segments of the signal.

↓ stationary

The Windowed Fourier Transform

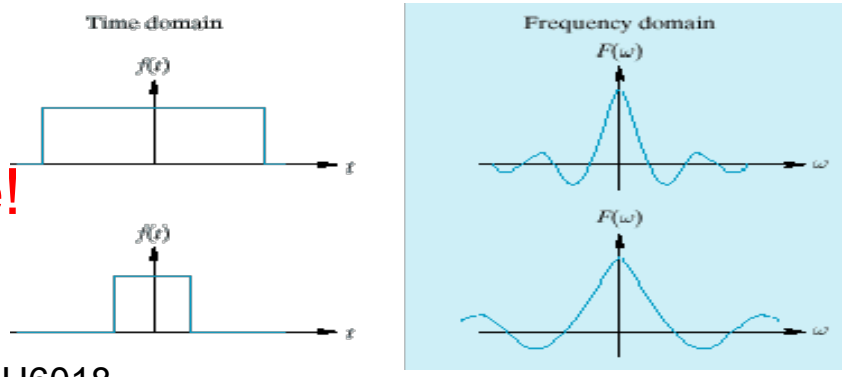
- We see that the basis function for the STFT is the product of the window function and the complex exponential of the FT.
- This is then multiplied by the signal to be transformed in order to perform the STFT.

$$STFT(t,\omega) = \int_{-\infty}^{\infty} s(\tau)\gamma^*(\tau - t)e^{-j\omega\tau}\,d\tau$$

About the Shape of the Window

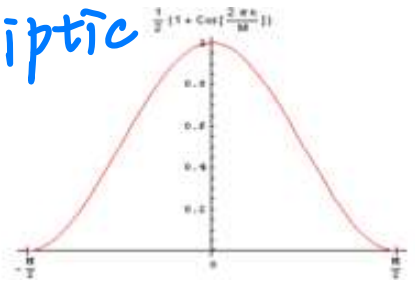
- If the window was rectangular, there would be discontinuities at the edges of the window, resulting in additional frequency components in the FT causing distortion of the spectrum.
- It would also require to be precisely translated so as to avoid overlap or gaps.
- Multiplication in the time domain is equivalent to convolution in the frequency domain, and the energy in the spectrum of the rectangular window is spread over its sinc function in frequency.
- So a rectangular window would be a very poor choice!

矩形窗 ×



About the Shape of the Window

- A better shape of window would be "bell" shaped, such as Hamming, Hann, Gaussian etc.



- The energy in the window is then concentrated near the middle and tapers towards the edges.
- This allows the window to overlap when moved and minimises distortion.
- The frequency spectrum of the "bell" shaped window is concentrated in the centre (approximating to an impulse in frequency).

Short Time Fourier Transform: Summary

- A window function $\gamma(t)$ is introduced.
- The transform is

$$STFT(t,\omega) = \int_{-\infty}^{\infty} s(\tau)\gamma^*(\tau - t)e^{-j\omega t}d\tau$$

- Compare with the Fourier transform $S(\omega) = \int_{-\infty}^{\infty} s(t)\,\mathrm{e}^{-j\omega t}\,dt$

- For each time t, there is a Fourier transform. We obtain a time-frequency representation of the signal.
- Typically, we compute and plot the **Spectrogram**

(a plot of the energy in the signal as a function of time and frequency)

$$\left|STFT(t,\omega)\right|^2$$

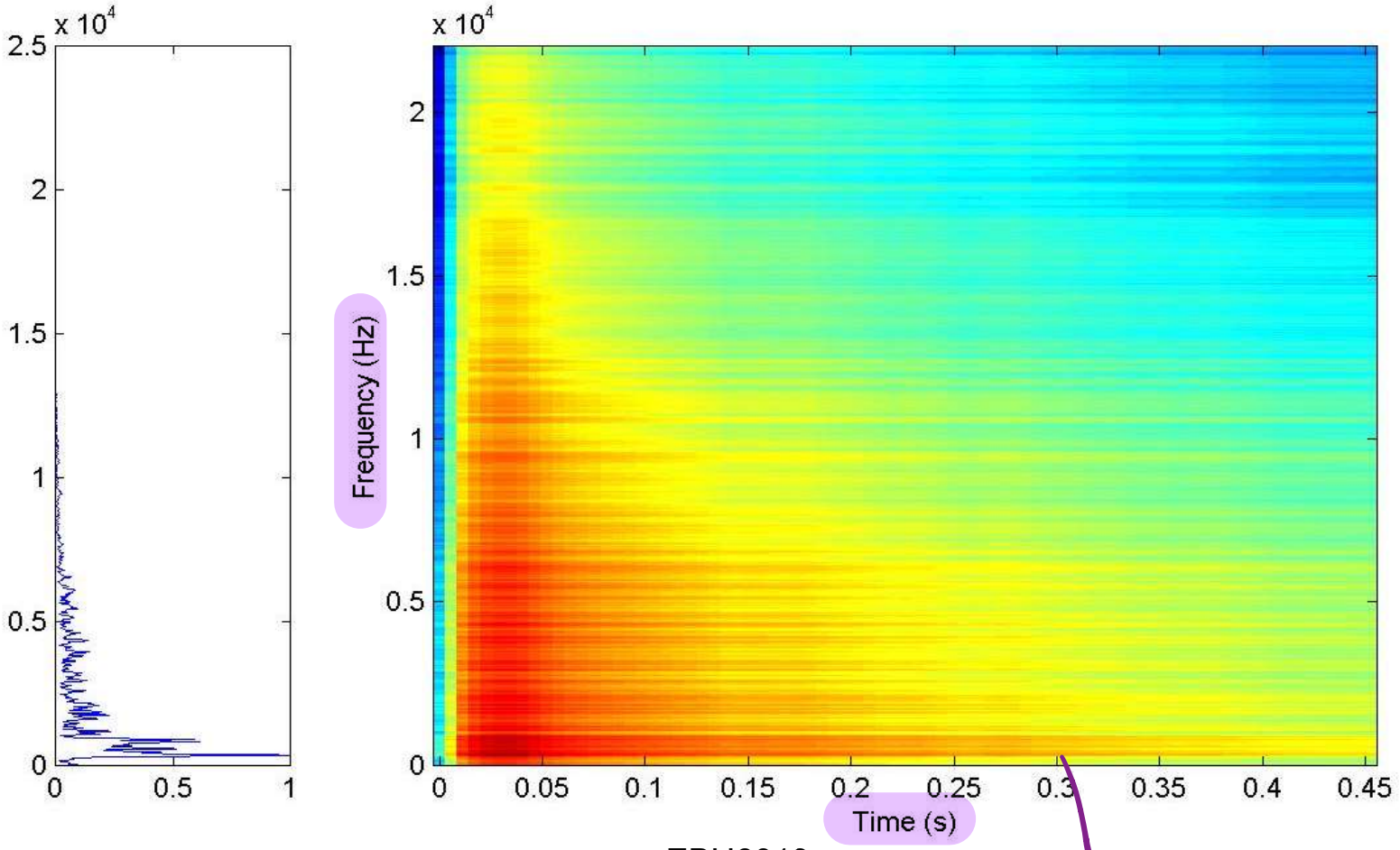
语谱图

$S(n,w) = |X(n,w)|^2$

to display the output from STFT

Spectrogram Example

Red means high energy, blue means low energy



3D plot

energy ↓

List the steps involved in performing a STFT.

The steps are:
1. Choose a window of finite length.
2. Place the window on top of the signal at t=0.
3. Truncate the signal using this window
4. Truncate the signal using this window
5. Incrementally shift the window to the right
6. Repeat from step 3

Gabor Expansion

- A discrete version of the STFT that uses a Gaussian function as the window.
- Represent a signal in 2 dimensions, with time and frequency as coordinates.
- Expand signal as a series of elementary functions.
- Constructed from a single building block by
  - Translation (in the time-domain)
  - modulation (translation in the frequency domain)
- Transform sampled at regular intervals.
- Intervals are labelled T in time and Ω in frequency.
- The transform divides the signal into regular windows

$$s(t) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} c_{m,n} h_{m,n}(t)$$

$$h_{m,n}(t) = h(t - mT) e^{jn\Omega t}$$

general form: $ae^{jbt}e^{-c(t-d)^2}$; simple form: e^{-at^2}

Choice of elementary function

this specific STFT: Gabor transform

- The most useful window function is a Gaussian window:

normalized form

$$h(t) = g(t) = \sqrt[4]{\alpha/\pi}\, e^{-\alpha t^2/2}$$

$$\Delta t \Delta \omega = \frac{1}{\sqrt{\alpha}} \frac{\sqrt{\alpha}}{2} = 1/2$$

最优情况

因为它的FT就是他本身

- Optimally concentrated in the time frequency domain.

优

- This function gives the best resolution & minimum distortion

因为满足 $\Delta t \Delta \omega = \frac{1}{2}$

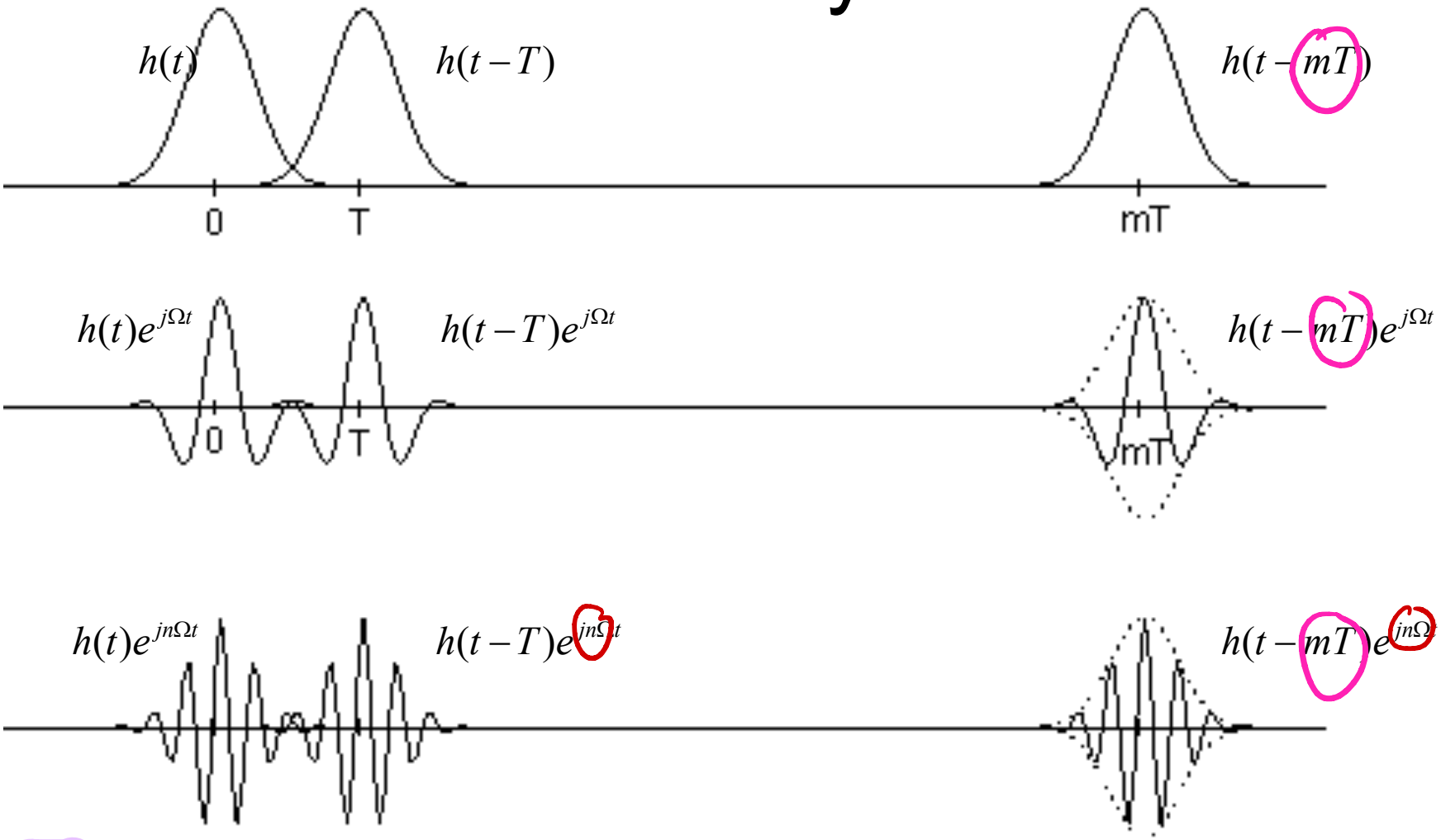
缺

- However, this does not produce orthogonal basis functions.

so it's necessary to find dual functions

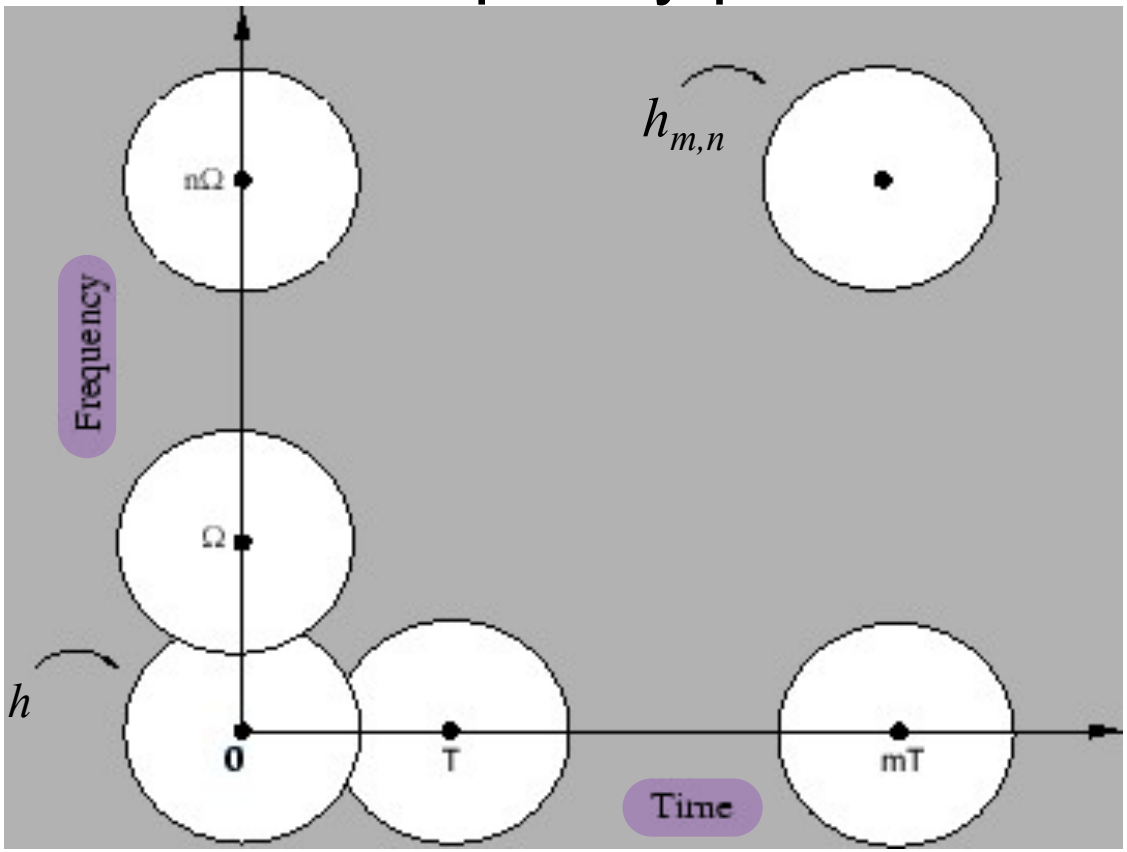
- Unlike Fourier transform, Gabor transform is not an orthogonal basis (and often not a tight frame).
- Finding the dual functions is typically not straightforward!

Gabor Elementary Functions



$h_{m,n}$ are shifted and modulated copies of a single building block h.
T denotes the time shift parameter.
Ω denotes the frequency shift parameter.
Each $h_{m,n}$ has an envelope of the shape of h.
Only the real part of the functions is shown.

Time-frequency plane



- $h_{m,n}$ are obtained by shifting h along a lattice in the *time-frequency plane*.
- If h and its Fourier transform are centred at the origin, then $h_{m,n}$ is centred at $(mT,n\Omega)$ in the time-frequency plane.
- Each $h_{m,n}$ essentially occupies a certain area in the time-frequency plane.
- Each expansion coefficient $c_{m,n}$, associated to a certain area of the time-frequency plane via $h_{m,n}$ represents one *quantum of information*.
- For properly chosen shift parameters the $h_{m,n}$ cover the time-frequency plane.

hm,n
Cm,n
信息
area of the plane
量子

Gabor frame properties

- Gabor frame operator

$$S = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \langle f, g_{m,n} \rangle g_{m,n}$$

- Commutes with translation

$$S(t - mT) = S(t) - mT$$

- Commutes with modulation

$$S(te^{jn\Omega t}) = S(t)e^{jn\Omega t}$$

- So the dual frame can be found through translation and modulation as well.

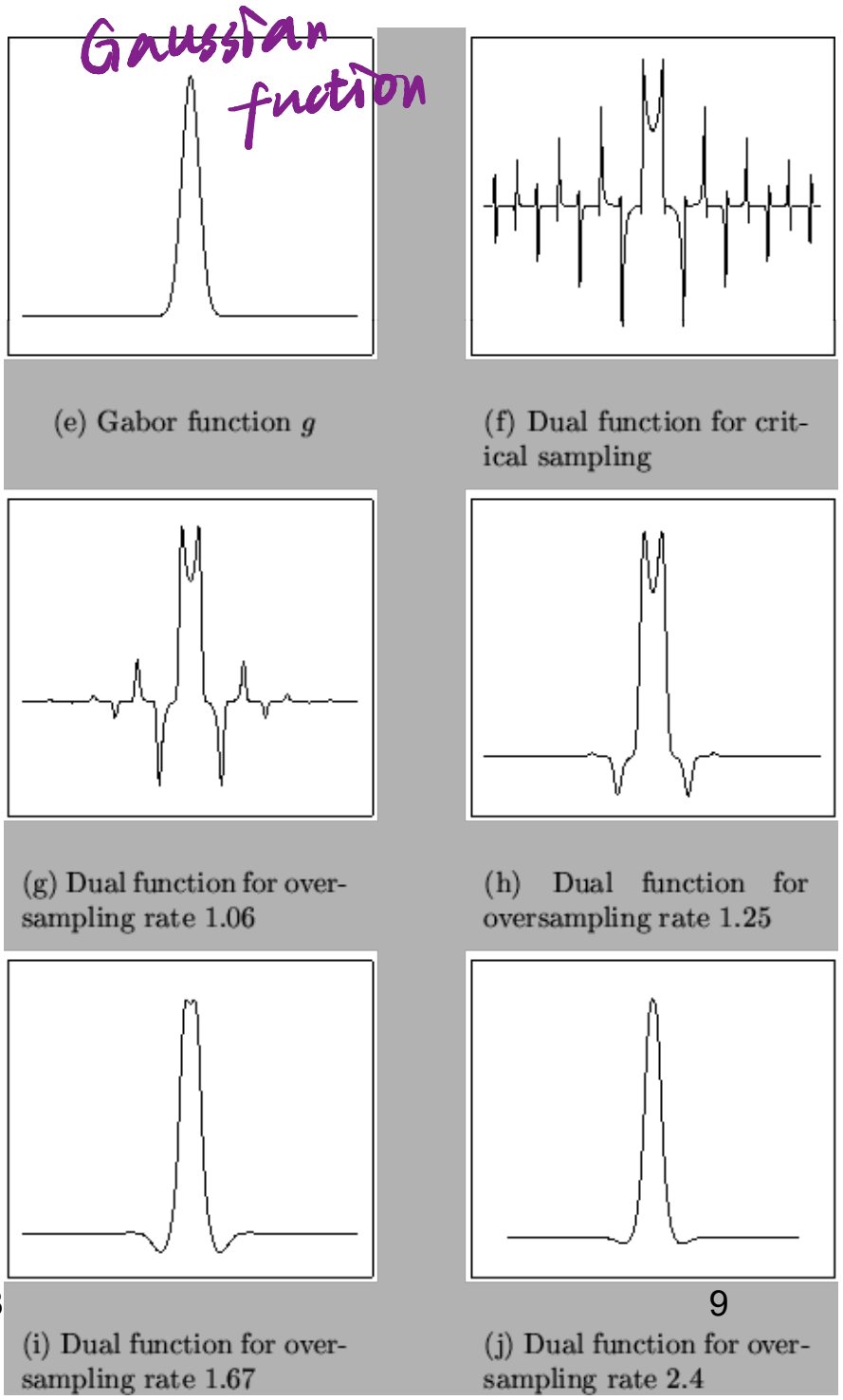
$$\gamma_{m,n} = S^{-1} g_{m,n} = S^{-1} g(t - mT)e^{jn\Omega t} = \gamma(t - mT)e^{jn\Omega t}$$

- Elements of a frame are in general linearly dependent.
- Many choices for the coefficients $c_{m,n}$ and even different choices of γ are possible.
- Coefficients determined by the dual frame are the most economical ones

Similar to Nyquist density for sampling and reconstruction of bandlimited functions in Shannon's Sampling Theorem.

Classify Gabor systems according to the corresponding sampling density of the time-frequency lattice:

- ***Oversampling*- Frames with excellent time-frequency localization properties exist**
  - **Gaussian with appropriate oversampling rate**
- ***critical sampling*- Frames and orthonormal bases possible, but without good time-frequency localization**
- ***Undersampling*- Any Gabor family will be incomplete. Cannot have a frame.**



(e) Gabor function g (f) Dual function for critical sampling

(g) Dual function for oversampling rate 1.06 (h) Dual function for oversampling rate 1.25

(i) Dual function for oversampling rate 1.67 (j) Dual function for oversampling rate 2.4

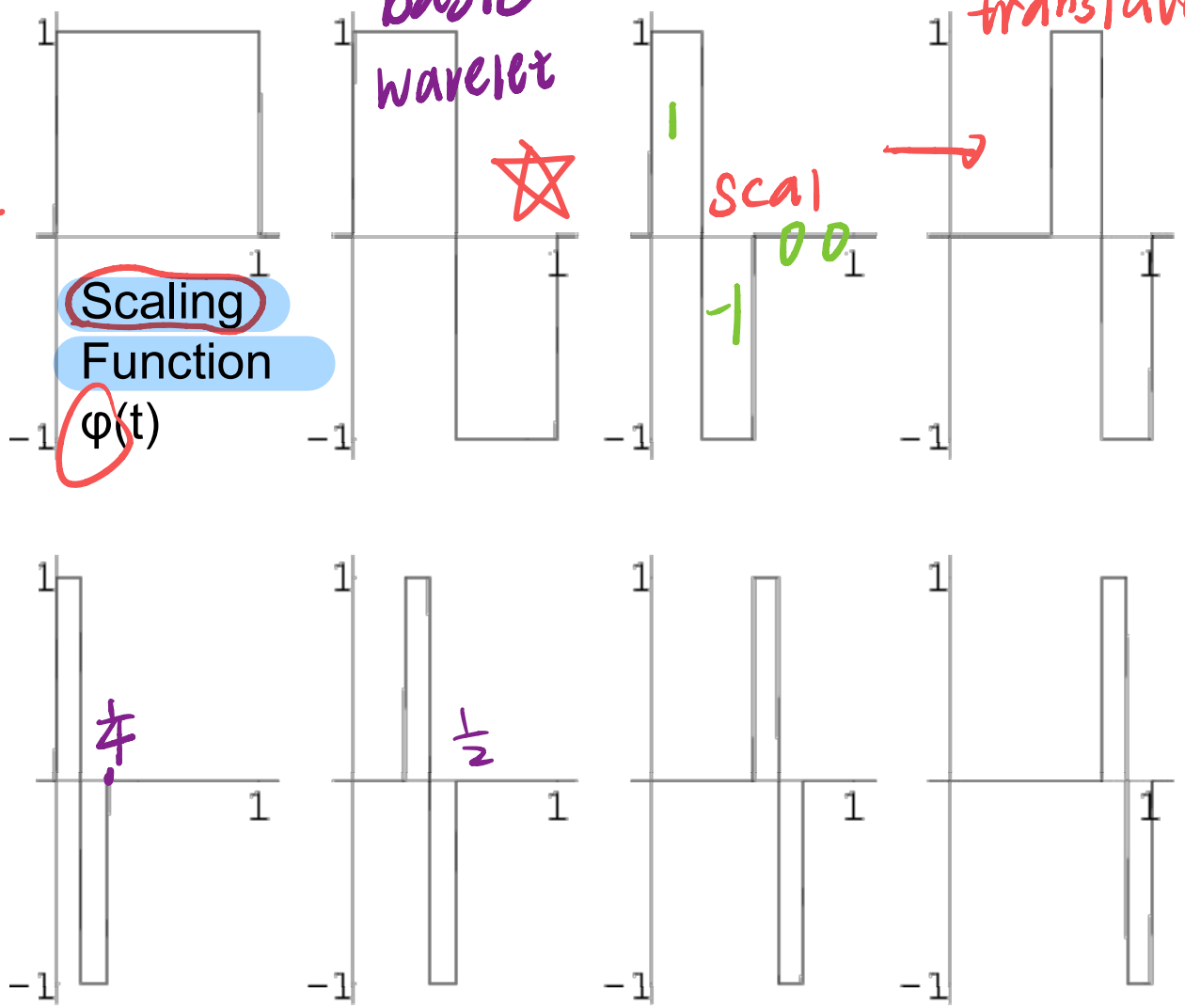
Haar Functions

- The Haar sequence of functions was first proposed by Alfred Haar in 1909 as an example of an orthonormal system.
- Although the concept of wavelets did not exist at that time, they can be considered to be the simplest wavelet function.
- Wavelet functions can be used as basis functions in the analysis of signals, similar to Fourier analysis but with the advantage of giving time-frequency analysis.
- Although Haar Functions are not continuous, they have some useful properties.

Wavelet Functions

- Each wavelet function forms a set of functions, which have a SCALING FUNCTION and a WAVELET FUNCTION
- These functions can be SCALED (i.e. Compressed) and TRANSLATED, forming a family of functions
- The Scaling is DYADIC (power of 2)
- The basic wavelet function is often called the MOTHER WAVELET, and the scaled and translated functions called DAUGHTER WAVELETS

Haar Functions



Wavelet Function:

$$\psi(x) \equiv \begin{cases} 1 & 0 \le x \le \frac{1}{2} \\ -1 & \frac{1}{2} \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\psi_{jk}(x) \equiv \psi(2^j x - k),$$

$\phi_{00}=\phi(x)$

$\psi_{00}=\psi(x)$ no skilled & translated

$\psi_{10}=\psi(2x)$

$\psi_{11}=\psi(2x-1)$

$\psi_{20}=\psi(4x)$

$\psi_{21}=\psi(4x-1)$ scal twice translate once

$\psi_{22}=\psi(4x-2)$

$\psi_{23}=\psi(4x-3)$

All Harr functions are defined in the time interval 0 to 1

Haar Function

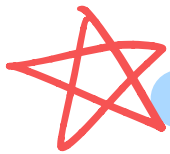
too simple.
cant analyse continuous / complexed information

- The basis functions of the Haar transform are known as the **Haar functions**
  - The Haar functions forms an orthogonal system 内积为0
  - The Haar functions are identified by scaling and translation daughter function
  - Compare it to the Fourier transform basis function, which only differs in frequency

- Haar functions are all rectangular-shaped functions

- Haar functions are defined on the interval [0, 1] and only take three values: {-1, 0, 1}

Orthogonal



And all members of the Haar Function family are mutually orthogonal.

For example, consider ψ_{10} and ψ_{11}

$\psi_{10} = [1, -1, 0, 0]$

$\psi_{11} = [0, 0, 1, -1]$

$$< \psi_{10}, \psi_{11} > = [1, -1, 0, 0] \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} = 0$$

Orthonormal

$$\psi_{jk}(x) = 2^{-j/2}\psi(2^j x - k)$$

For example, ψ_{10} = [1, -1, 0, 0]

Norm is $\sqrt{2}$

So normalised, $\psi_{10} = \frac{1}{\sqrt{2}}$[1, -1, 0, 0]

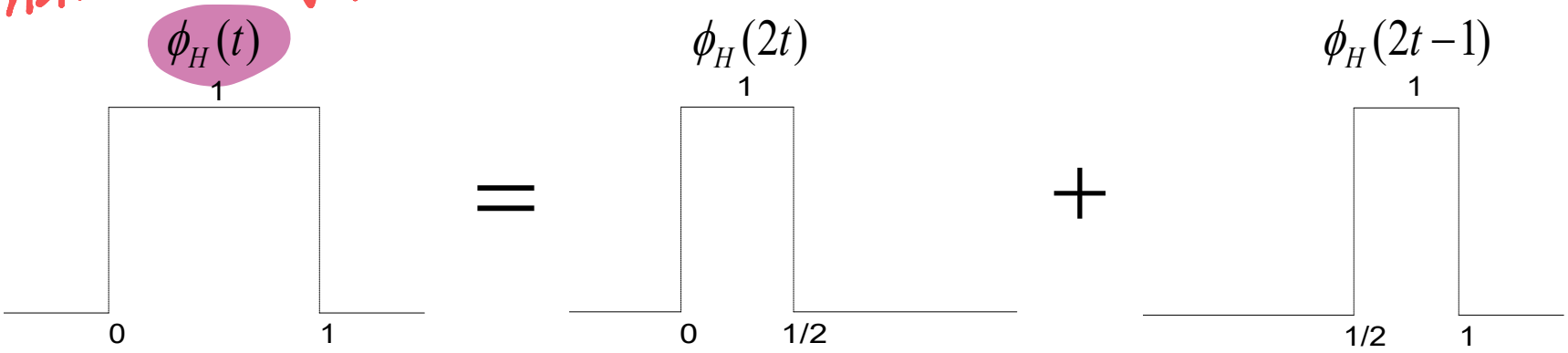
Feature Extraction：
if we use the Haar Matrix to perform a transform we can then find short duration signals of different frequencies located at different times in a signal
Performing a Haar Transform on signal could identify those features and the times they occurred.
correlation would not be very good，因为Haar function和特征信号形状不同，（所以应该找与特征信号相似的函数，which is Wavelet）

Two-Scale Relations

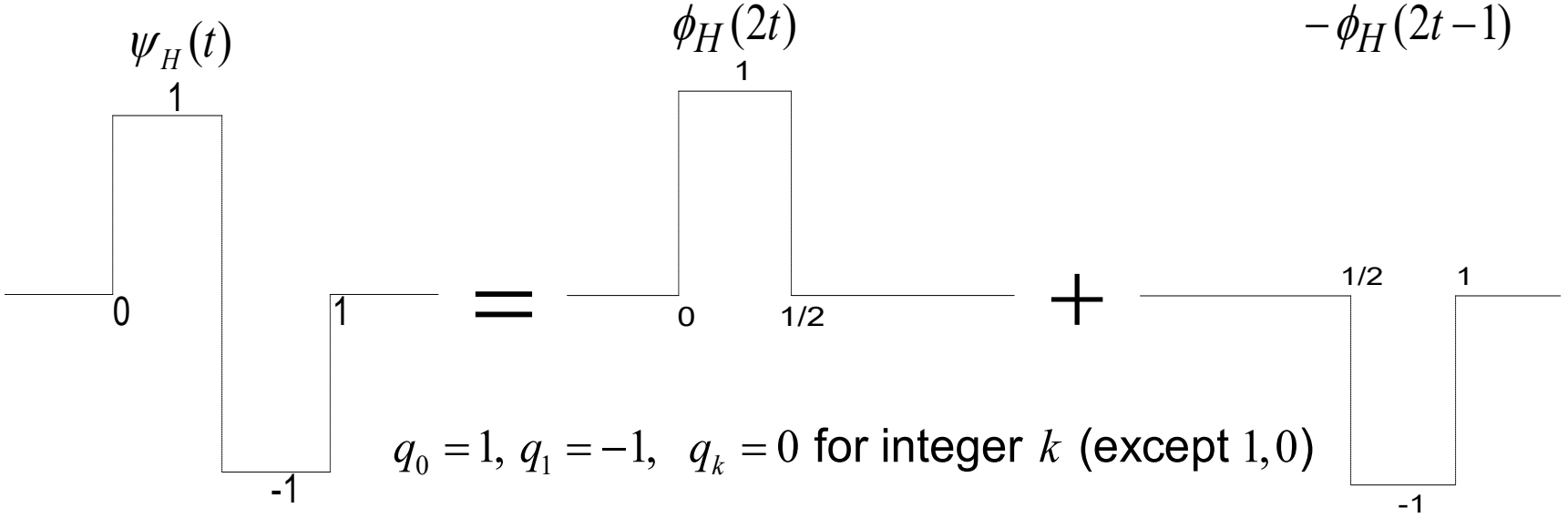
- A useful property of Haar Functions is that both the Scaling Function and the Wavelet Function can be constructed from scaled and translated versions of the Scaling Function only.
- This property can be used when Haar Functions are applied to Wavelet Transforms.
- In wavelet transform theory this property is called the "Two-Scale Relations"

Two-Scale Relations

haar scaling function



$p_0 = p_1 = 1, \quad p_k = 0$ for integer k (except $1, 0$)



$q_0 = 1, \; q_1 = -1, \quad q_k = 0$ for integer k (except $1, 0$)

Linear Piecewise Approximation

- Another useful property of Haar Functions is that they can be used to approximate any continuous real function by linear combinations of scaled and translated members of the family of Haar Functions.

- This principle is illustrated on the following slide:

Summary

Haar Functions are a relatively simple set of functions with some useful properties:

- They are orthogonal (and can be normalised)
- They can be used to approximate any continuous real function
- Both the Scaling Function and the Wavelet Function can be derived from the Scaling Function only
- The application to wavelet transforms will be discussed after the lecture on Wavelet Transforms

Haar Transform

use Haar Matrix formed through Haar Functions.

- The Haar transform was proposed in 1910 by a Hungarian mathematician Alfred Haar. It is one of the earliest transform functions proposed

- Haar transform provides time-frequency information, hence is suitable for non-stationary signals
  - Compare it to Fourier transform, which only provides frequency information

- Haar transform is the simplest of the wavelet transform

Haar Transform - Applications

- Haar transform is beneficial for image/signal compression
  - However, Haar transform is a lossy data compression
  - The advantage of Haar transform is that it can achieve high compression ratio versus lossless techniques

- Haar transform is also good for denoising 降噪
  - When signal is received after transmission over some distance, it is often distorted by noise.

- General procedure: After Haar transform is performed, a thresholding is used, i.e., any values of the transformed signal lie below the noise/compression threshold is set to 0. Then, inverse Haar transform is performed to obtain the denoised/compressed signal

Haar Transform - Quiz

Question 1

Which of the following is true?

a. Haar transform is a non-orthogonal transform — The Haar transform is an orthogonal transform
b. Haar transform has fixed basis functions — Correct ! The basis functions are independent of the signal
c. Haar transform is slow — The Haar transform is fast
d. Haar transform is complex-valued — The Haar transform is real-valued

Summary

- We have seen that a Haar Matrix can be constructed to perform Haar Transforms directly.
- The **Haar Transform is fast** because the matrix contains many zero terms and it is real (no complex terms).
- It can be used to **identify frequency components** in the signal to be analysed (fine detail).
- It can be used to **identify the trends in the input data** (approximations).
- It can be used for **compression** by reducing or eliminating the coefficients corresponding to high frequencies in the signal and then inverting the transform.

Applications of Wavelets

- Relatively new method of evaluating and processing signals
- Works on nonstationary data
- Two main applications are in feature extraction and trend analysis
- Useful in many types of applications
  - Pattern recognition
    - Biotech: distinguish normal from pathological membranes
    - Biometrics: facial/corneal/fingerprint recognition
  - Feature extraction
    - Metallurgy: characterization of rough surfaces
  - Trend detection:
    - Finance: exploring variation of stock prices
  - Perfect reconstruction
    - Communications: wireless channel signals
  - Video compression – JPEG 2000

The Wavelet

- Consider scaling and translating the function *y*

$$\psi(t) \rightarrow \psi\left(\frac{t-b}{a}\right)$$

- *a* determines the centre frequency.
- *b* determines the translation.
- Time frequency centre of $\psi\left((t-b)/a\right)$
  are *b* (time centre)
  and $\langle\omega\rangle/a$ (frequency centre) $\langle\omega\rangle$ is mean freq of ψ

- Daughter wavelets:

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}}\psi\left(\frac{t-b}{a}\right)$$

Mother Wavelet

Continuous Wavelet Transform

$$CWT(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} s(t)\psi^*\left(\frac{t-b}{a}\right) dt$$

Scale

Translation

$$= \int_{-\infty}^{\infty} s(t)\psi_{a,b}{}^*(t)dt = < s, \psi_{a,b} >$$

The wavelet function must be oscillatory.

- The continuous wavelet transform, $CWT(a,b)$ is a function of two real variables.
- Compare short-time Fourier Transform:

$$STFT(t,\omega) = \int_{-\infty}^{\infty} s(\tau)\gamma^*(\tau - t)e^{-j\omega t} d\tau$$

- Have $\psi_{a,b}{}^*(t)$ instead of $\gamma^*(\tau - t)e^{-j\omega t}$

CWT: Time-Frequency Analysis

- CWT provides a time-frequency as well as time-scale representation.

$$CWT(a,b) = TF(t = b, \omega = \langle \omega \rangle / a)$$

- We can define the *Scalogram*

$$SCAL(a,b) = |CWT(a,b)|^2$$

- Compare Spectrogram: $|STFT(t,\omega)|^2$

Scaling of a signal

Consider time-scaling a signal: $r(t) = s(t/\alpha)$

This changes Fourier Transform: $R(\omega) = \alpha S(\alpha\omega)$

So changes energy: $E_r = \int_{-\infty}^{\infty} |s(t/\alpha)|^2 \, dt = \int_{-\infty}^{\infty} |s(\tau)|^2 \, d(\tau\alpha) = \alpha E$

New centre freq:

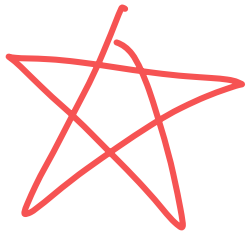
$$\langle\omega\rangle_R = \frac{1}{2\pi E_R} \int_{-\infty}^{\infty} \omega |R(\omega)|^2 \, d\omega$$

$$= \frac{1}{2\pi\alpha E} \int_{-\infty}^{\infty} \omega |\alpha S(\alpha\omega)|^2 \, d\omega \qquad R(\omega)=\alpha S(\alpha\omega)$$

$$= \frac{1}{2\pi\alpha E} \int_{-\infty}^{\infty} \frac{\Omega}{\alpha} |\alpha S(\Omega)|^2 \, d\frac{\Omega}{\alpha} \qquad \Omega=\alpha\omega$$

$$= \frac{1}{2\pi\alpha E} \int_{-\infty}^{\infty} \Omega |S(\Omega)|^2 \, d\Omega = \frac{\langle\omega\rangle}{\alpha}$$

Scaled centre freq



Scaling (cont)

New frequency width:

$$\Delta_\omega^2(R) = \frac{1}{2\pi E_R}\int_{-\infty}^{\infty}\omega^2\left|R(\omega)\right|^2 d\omega - \left\langle\omega\right\rangle_R^{\,2}$$

$$= \frac{1}{2\pi\alpha E}\int_{-\infty}^{\infty}\omega^2\left|\alpha S(\alpha\omega)\right|^2 d\omega - \left(\left\langle\omega\right\rangle/\alpha\right)^2$$

$$= \frac{1}{2\pi\alpha E}\int_{-\infty}^{\infty}(\Omega/\alpha)^2\left|\alpha S(\Omega)\right|^2 d\frac{\Omega}{\alpha} - \left\langle\omega\right\rangle^2/\alpha^2$$

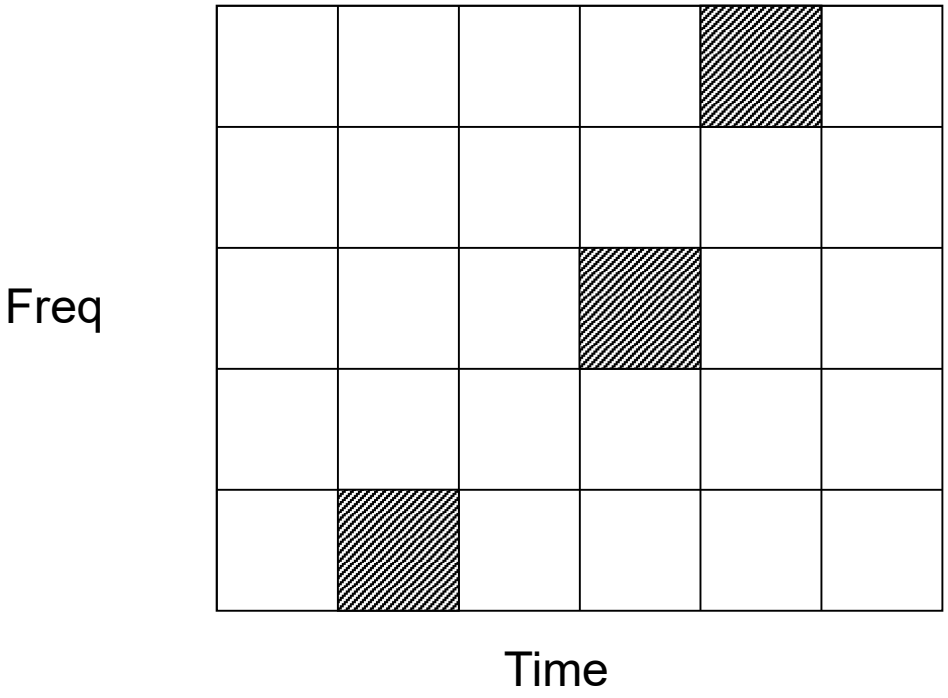
$$= \frac{1}{2\pi\alpha^2 E}\int_{-\infty}^{\infty}\Omega^2\left|S(\Omega)\right|^2 d\Omega - \left\langle\omega\right\rangle^2/\alpha^2$$

$$= \frac{\Delta_\omega^2(S)}{\alpha^2}$$

Scaled frequency resolution

Time-Freq Partition: STFT

FT: Equal time and frequency resolution



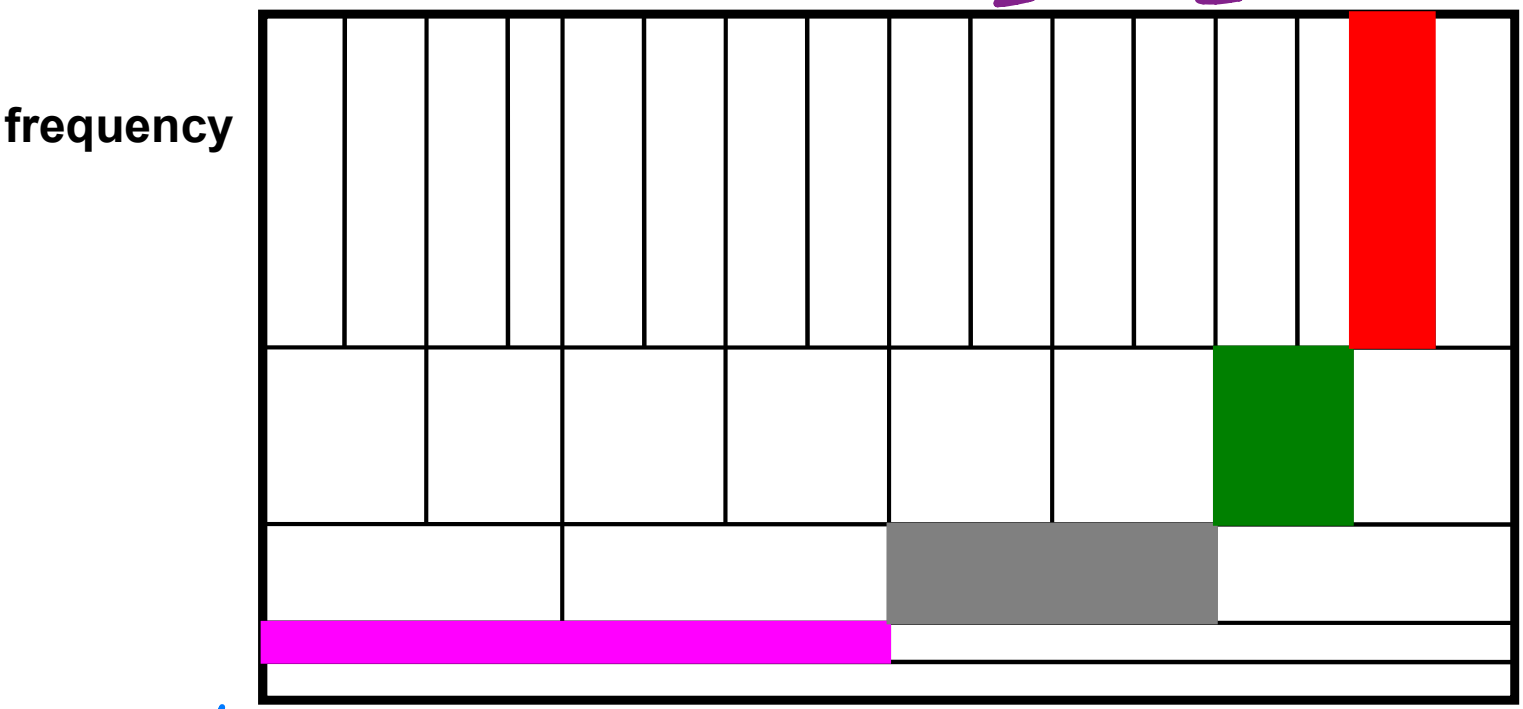
the window is fixed width, so that Δt is constant. Because of the uncertainty principle, $\Delta t \cdot \Delta f = k$, therefore Δf is also fixed. So the resolution is constant for all time and frequency.

(WT: Logarithmic scale of frequency resolution)

窗大小一旦选定无法改变，对于给定窗 → fixed resolution. 若要改resolution → 重新选择窗.

Partition of the time-frequency plane

- High scale (low frequency)
  - large window size, better frequency resolution
- Low scale (high frequency)
  - small window size, better time resolution.



time-frequency tiling of wavelet transform

the wavelet function is scalable and therefore can be compressed resulting in Δt being variable. This results in Δf being inversely proportional to Δt.

Inverse CWT: The Admissability Criterion

- We can construct an Inverse FT to reconstruct $s(t)$
  Can we do the same for CWT?
- Yes: provided that the *Admissibility Condition* is satisfied:

$$C_\Psi = \int_{-\infty}^{\infty} \frac{|\Psi(\omega)|^2}{|\omega|} d\omega < \infty$$

has to be finite

where $\Psi(\omega) = \int_{-\infty}^{\infty} \psi(t)\, e^{-j\omega t}\, dt$ is the Fourier Transform of $\psi(t)$

Reconstruction:

$$s(t) = \frac{1}{C_\Psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{a^2} CWT(a,b)\psi_{a,b}(t) da db$$

Admissibility Condition (cont)

- Square of the Fourier transform must decay faster than 1/*w*.
- Admissibility is measure of signal's band-limitedness.
- Admissibility implies zero average:

$$\Psi(0) = \int_{-\infty}^{\infty} \psi(t)\, \mathrm{e}^{-j0t}\, dt = \int_{-\infty}^{\infty} \psi(t) dt = 0$$

because otherwise $\dfrac{|\Psi(\omega)|^2}{|\omega|} \to \infty$ as $\omega \to 0$

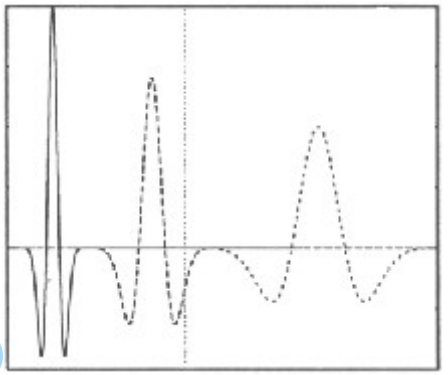
CWT versus STFT

CWT: Variable time-frequency resolution

$$CWT(a,b) =< s, \psi_{a,b} >$$

Scale, Translation

Different width;
Same no of cycles

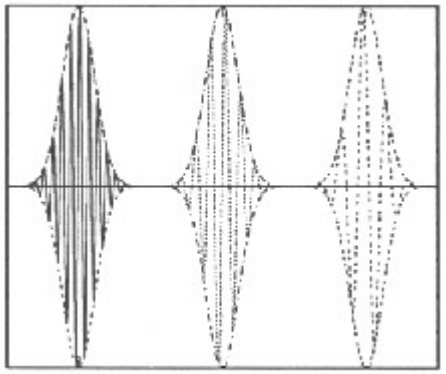


STFT: Constant time-frequency resolution

$$STFT(u,b) =< s, g_{u,b} >$$

Frequency, Translation

Same width;
Different no of cycles



Comparison of STFT and CWT

- Similarities:
  - signal is multiplied by a function, and the transform is computed separately for different segments of signals.
  - can be written in inner product form

$$STFT(b,\omega) = \left\langle s(t), \gamma(t-b)e^{j\omega t} \right\rangle \qquad CWT(b,a) = \left\langle s(t), \frac{1}{\sqrt{a}}\psi\left(\frac{t-b}{a}\right) \right\rangle$$

  - Time-frequency window area remains constant.

- Difference:
  - Fixed time duration and freq bandwidths of $\gamma(t)$
  - Variable time duration and bandwidth of $\psi(t)$

Problems with CWT

Redundancy (because continuous)

- Basis functions for CWT are shifted and scaled versions of each other. Usually do not form an orthonormal base.

Infinite solution space

- The result holds an infinite number of wavelets:
  hard to solve and hard to find the desired results out of the transformed data.

Efficiency

like FFT for DFT

- Most transforms cannot be solved analytically. Solutions have to be calculated numerically: time-consuming.
  Must find efficient algorithms.

Solution?

- *Multiresolution Analysis*

Wavelet transforms can be used for many applications. Briefly describe each of the following:
i) Resolving a signal into its component sinusoids
ii) De-noising
iii) Detecting discontinuities
iv) Detecting self-similarity
v) Compressing images.

Resolving a signal into its frequency component: MRA separates a waveform into its coarse and fine detail till only the average value is left. [1 mark] The detail that has been separated out are the frequencies in the signal with the highest ones being separated out first [1 mark]
De-noising: if all the noise is high frequency it can be removed by compression or by discarding the first levels of detail in the MRA [l mark]. However, if the noise is broadband or has different components at different frequencies then denoising can be carried out by identifying the features in the signal that we are looking for and ignoring the rest. This can be achieved by using wavelet function of the same shape as the features t be identified [l mark]
Detecting discontinuities: if the discontinuity is between bursts of different frequencies, then at some level of decomposition the discontinuity will be obvious because the higher frequency will disappear from the coarse level [l mark]. It will also detect an abrupt change in the rate of change [1 mark].
Detecting self-similarity: wavelets can be used to detect fractal signals (i.e. similar to itself at different scales) then the wavelet coefficients will also be similar at different scales [l mark]. This generates a set of characteristic lines on the scalogram [1 mark].
Compression of images: This is carried out by performing the MRA and then setting a threshold value [1 mark]. values lower than the threshold value are discarded resulting in compression [l mark].

MRA

CWT → DWT

Tackling the Redundancy Problem

- Because the CWT(a,b) is a continuous transform, it encodes more than enough information to reconstruct the signal. A sampled version is sufficient.
- a is scaling factor. b is translation term.
- Let us sample in a dyadic (power of 2) grid:

$$a = 2^{-m} \qquad b = n2^{-m}$$

$$CWT(2^{-m}, n2^{-m}) = \frac{1}{\sqrt{|2^{-m}|}} \int_{-\infty}^{\infty} s(t)\psi^*\left(\frac{t - n2^{-m}}{2^{-m}}\right)dt$$

basic function; b; a; normalization factor

$$d_{m,n} = CWT(2^{-m}, n2^{-m}) = \int_{-\infty}^{\infty} s(t)\psi^*_{m,n}dt$$

where $\psi_{m,n}(t) = 2^{m/2}\psi(2^m t - n)$

Discrete (sampled) Transform

If the set of functions $\psi_{m,n}$ forms a frame, then we can recover s(t)

$$s(t) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} d_{m,n} \hat{\psi}_{m,n}(t)$$

where $\hat{\psi}_{m,n}(t)$ forms the dual frame

For the orthonormal frame, we have $\hat{\psi}_{m,n}(t) = \psi_{m,n}(t)$

So we get the *wavelet series* (transform and inverse):

$$d_{m,n} = \int_{-\infty}^{\infty} s(t) \psi^*_{m,n} dt$$

$$s(t) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} d_{m,n} \psi_{m,n}(t)$$

reconstruction

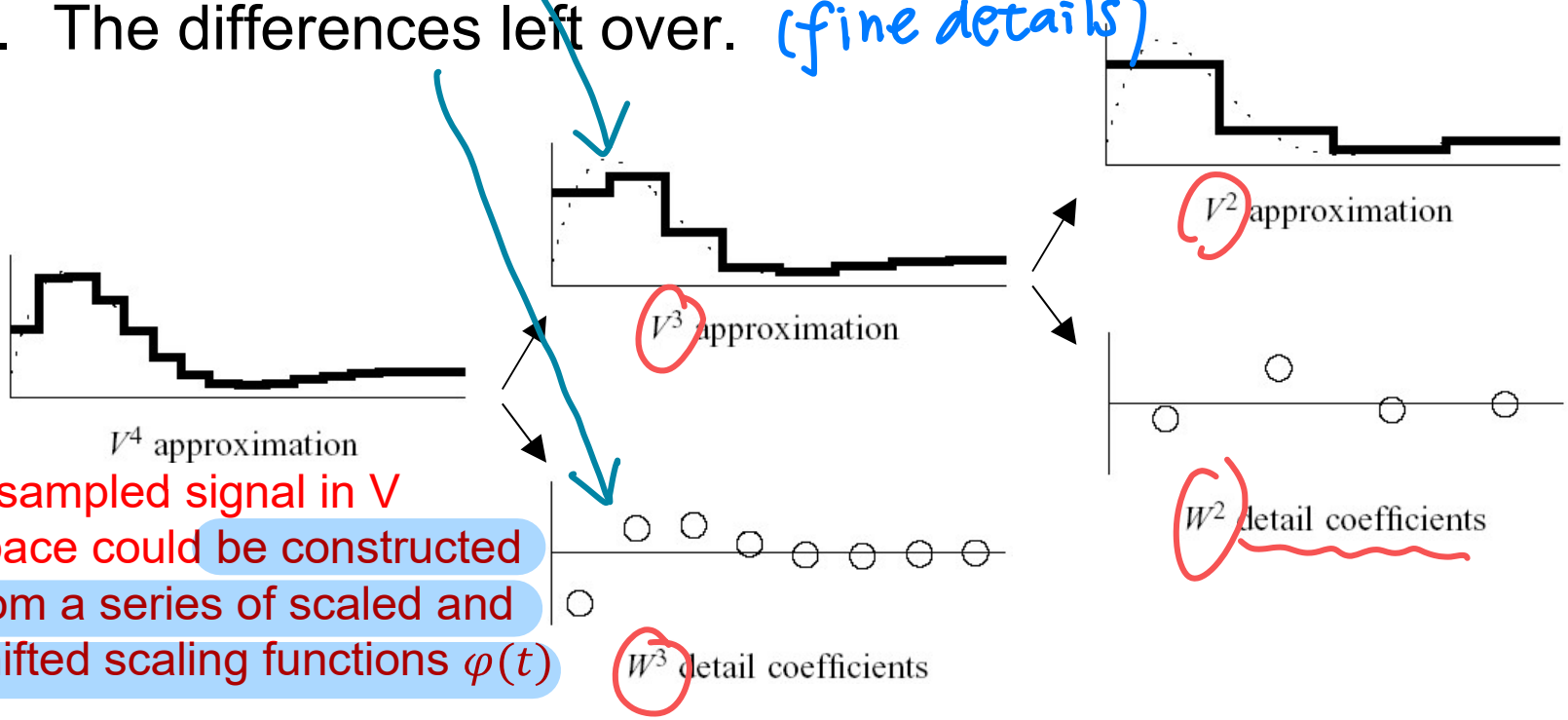
So: wavelet series is not redundant (finite number of coefficients)

Piecewise Approximation

MRA's

Basic Concept: Decompose a fine-resolution signal into

1. A coarse-resolution version of the signal, and
2. The differences left over. (fine details)



A sampled signal in V space could be constructed from a series of scaled and shifted scaling functions $\varphi(t)$

So: what about wavelets? Theory coming next...

Multiresolution Analysis

- If we remove the fine detail from the signal in space V^4
- and put that in space W^3 then the approximate signal left in V^3 and the detail in W^3 have no elements in common.
- So the signal in V space and the signal in W space at a given level of decomposition are orthogonal.
- Similarly for V^2 and W^2, etc.
- Starting at level 4 is arbitrary.

Multiresolution Analysis (cont)

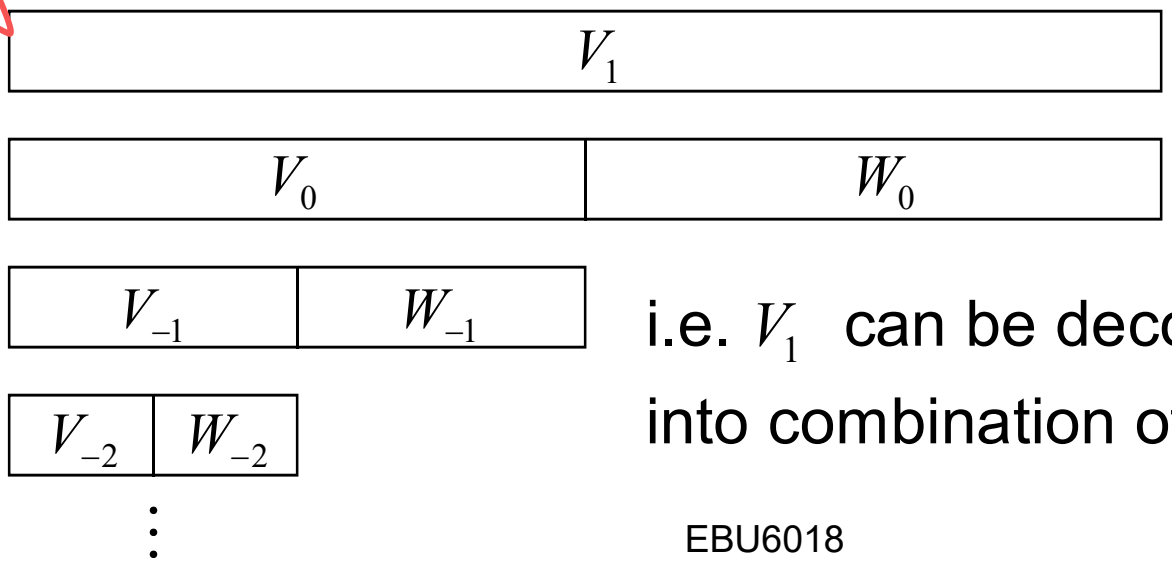
Subspaces $\{V_j\}$ are nested

while subspaces $\{W_j\}$ are mutually orthogonal.

Consequently,

$$\begin{cases} V_j \cap V_l = V_j & l > j; \\ W_j \cap W_l = \{0\} & j \neq l; \\ V_j \cap W_l = \{0\} & j \leq l. \end{cases}$$

$\vdots$



$\vdots$

i.e. V_1 can be decomposed into combination of V_0 and W_0, etc.

Multiresolution Analysis (cont.)

If the Scaling function and the Wavelet function are orthogonal, as they are for the Haar Function, then we can say that because V and W are orthogonal:

Subspaces $\{W_j\}$ are generated by *wavelets* $\psi(t)$

And $\{V_j\}$ is generated by scaling functions $\phi(t)$

In other words, any $f_j(t) \in V_j$

can be written as $f_j(t) = \sum_k c_k^j \phi(2^j t - k),$

And……

the wavelet function gives the fine details in the signal

the scaling function gives the coarse detail.

Multiresolution Analysis (cont.)

Subspaces $\{V_j\}$ are called "approximation subspaces" and $\{W_j\}$ are the "wavelet subspaces".

At any level j, V_j contains the smooth part and W_j contains the "details" of the original function.

$$\begin{cases} j\uparrow \Rightarrow f_j(t), \text{ a finer approximation of } f(t), \\ j\downarrow \Rightarrow f_j(t), \text{ a coarser approximation of } f(t). \end{cases}$$

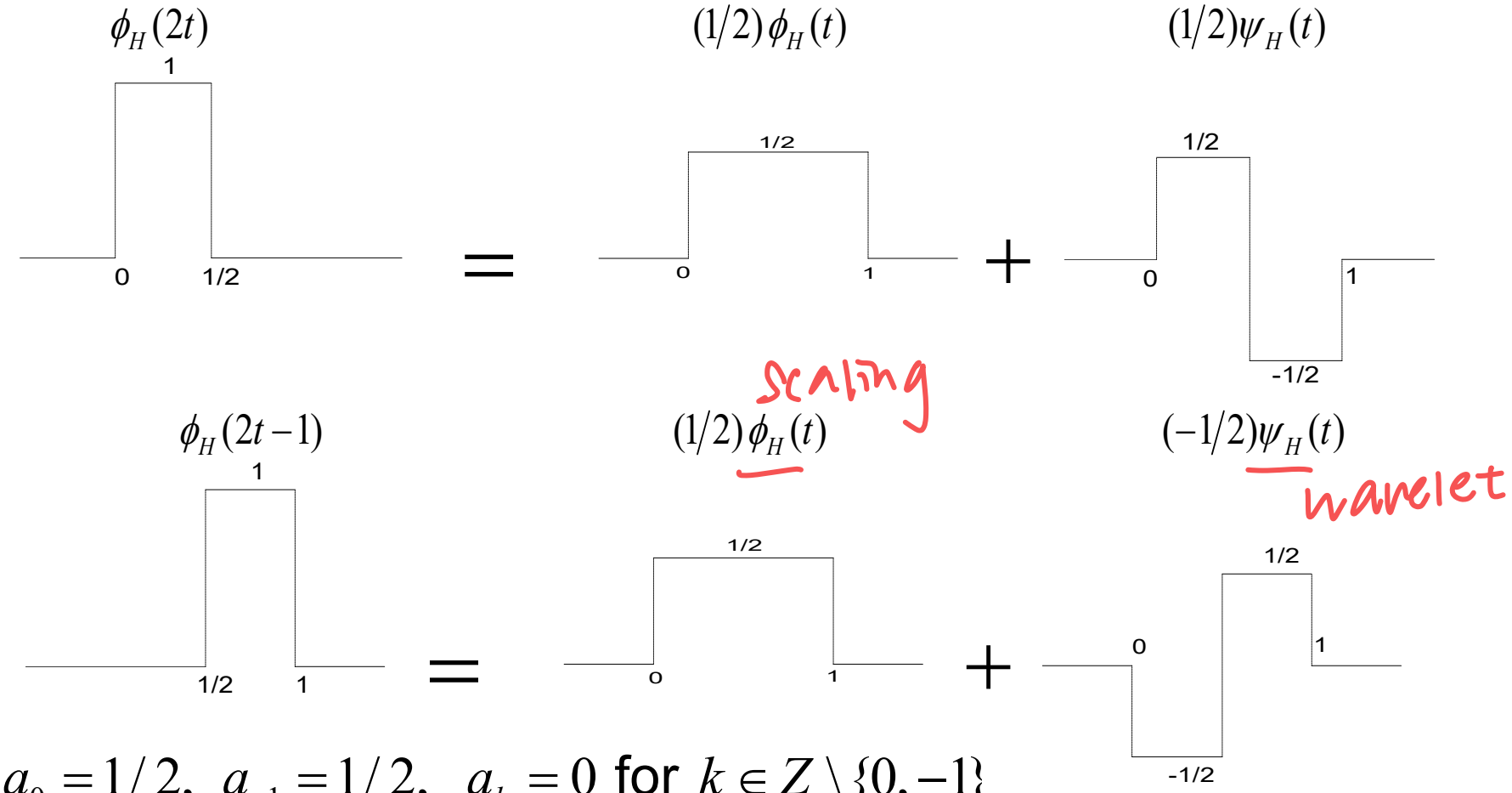
less details

Decomposition

The approximations can be derived as a series of scaled and shifted scaling functions, each with a coefficient (same as a linear piecewise approximation)

The fine detail can be derived as a series of scaled and shifted wavelet functions, each with a coefficient

Example: (H: Haar)



$$a_0 = 1/2,\ a_{-1} = 1/2,\ a_k = 0 \text{ for } k \in Z \setminus \{0,-1\}$$

$$b_0 = 1/2,\ b_{-1} = -1/2,\ b_k = 0 \text{ for } k \in Z \setminus \{0,-1\}$$

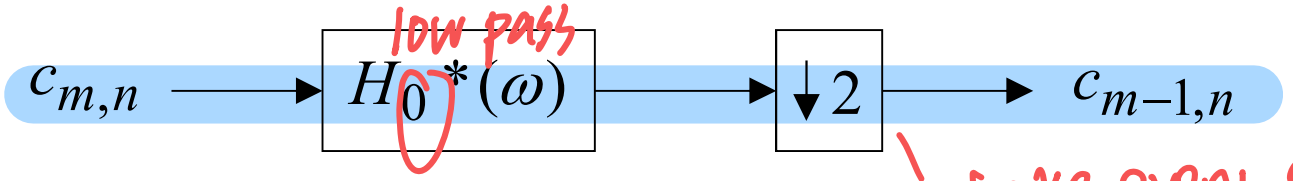
WTFB

Filtering and downsampling

Given this eqn $c_{m-1,n} = \sqrt{2}\sum_i h_0[i-2n]c_{m,i}$

once $c_{m,n}$ is known, we can compute $c_{k,n}$ for $k < m$,

using a low pass filter $H_0{}^*(\omega)$ and downsampling $2n = i$.

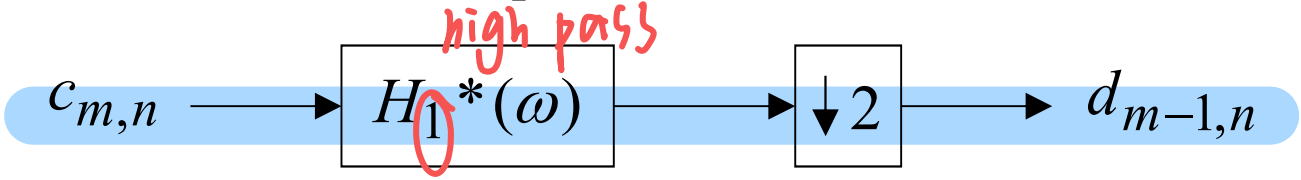


Low-pass filter for approximations

take every second value

Similarly, we can show $d_{m-1,n} = \sqrt{2}\sum_i h_1[i-2n]c_{m,i}$ ***

i.e. a high pass filter $H_1{}^*(\omega)$ and downsampling.



High-pass filter for fine detail

Recursive computation of coeffs

Dilation equation $\phi(t) = \sum_k p_k \phi(at - k)$ for integer k

From MRA, if φ(t/2) ∈ V_{-1} then φ(t/2) ∈ V_0

Let us define

$c_{m,n} \equiv s[n] \equiv s(t)\big|_{t=2^{-m}n}$ for large m.

And define a filter $h_0[n]$ Such that we get the refinement equation:

Using the dilation equation $\phi(t/2) = 2\sum_n h_0[n]\phi(t-n)$ we get

证明

$$c_{m-1,n} = \int_{-\infty}^{\infty} s(t)\phi^*_{m-1,n}(t)dt = 2^{(m-1)/2}\int_{-\infty}^{\infty} s(t)\phi^*\left(\frac{2^m t - 2n}{2}\right)dt$$

$$= 2^{(m-1)/2}\int_{-\infty}^{\infty} s(t)2\sum_i h_0[i]\phi^*(2^m t - 2n - i)dt$$

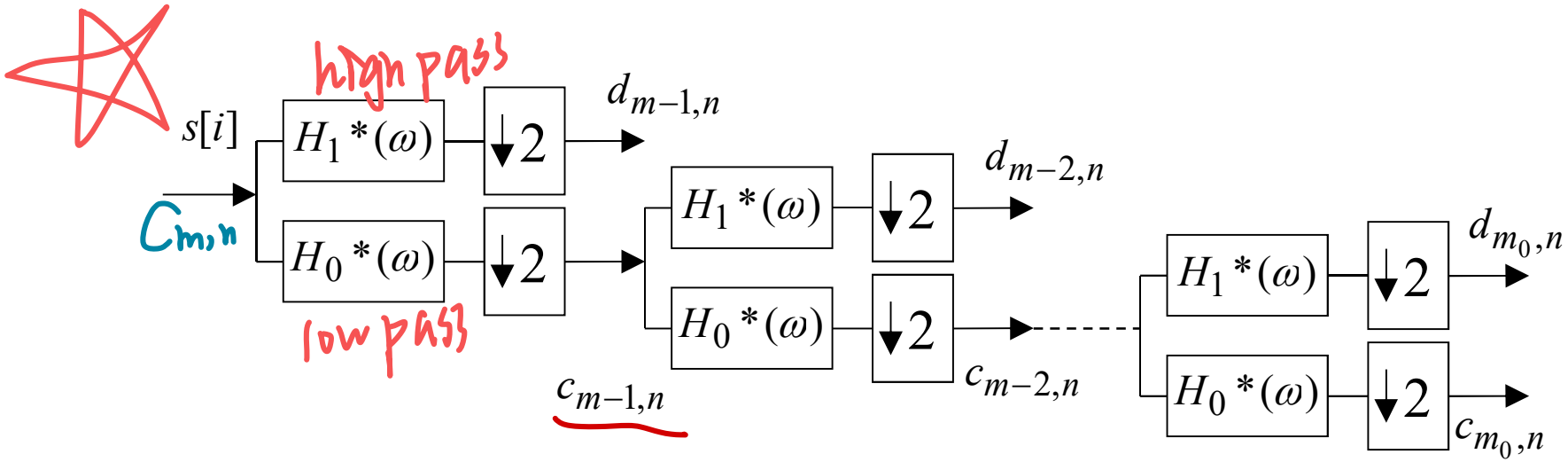
$$= \sqrt{2}\sum_i h_0[i]\int_{-\infty}^{\infty} s(t)\phi^*_{m,2n+i}(t)dt = \sqrt{2}\sum_i h_0[i]c_{m,2n+i}$$

$c_{m-1,n}$ h_1 $c_{m,i}$

i.e. $c_{m-1,n} = \sqrt{2}\sum_i h_0[i-2n]c_{m,i}$ ***

$c_{m-1,n}$ 的求法

Filter Bank for Wavelet Series Coeffs



So for discrete-time samples $s[i]$, can compute wavelet transform directly by applying filter banks.

No need to compute the mother wavelet $\psi(t)$.

a) In the wavelet transform, the scaling function coefficients $c_{m,n}$ and wavelet series coefficients $d_{m,n}$ can be calculated recursively according to the following equations:

$$c_{m-1,n} = \sqrt{2}\sum_i h_0[i-2n]c_{m,i}$$
$$d_{m-1,n} = \sqrt{2}\sum_i h_1[i-2n]c_{m,i}$$

Explain how this can be interpreted in terms of filtering and downsampling, and hence leads to the concept of an *analysis filterbank*. Sketch a diagram to illustrate this filterbank. [6 marks]

Answer:

a) If $c_{m,i}$ are the scaling function coefficients at level m, we can calculate the scaling function coefficients and wavelet series coefficients $d_{m-1,n}$ at level m-1 recursively from these. [2 marks]

The signals h_0 and h_1 are (time-reversed) low-pass and high-pass filters. [1 mark]
Steps of 1 in n correspond to steps of 2 in i, corresponding to downsampling by a factor of 2 from n to i. [1 mark]

Therefore these equations represent filtering followed by downsampling, as shown in the diagram:

Solutions

A signal can be filtered into its coarse and fine detail (analyzed) by using these wavelet function and scaling function. Sketch and explain the block diagram for one stage of a dyadic wavelet transform. How could this process be used for a non-dyadic sequence?

Dyadic means that the number of values in the sequence being transformed is a power of 2 [1 mark]. The inputs to this stage are the coarse details from the previous stage [1 mark]. H0 is a low pass filter to obtain the next level of coarse detail [1 mark]. H1 is a high pass filter to obtain the next level of fine detail [1 mark]. Downsampling is carried out so that at each level there is half the number of values of each as at the previous stage [1 mark]. If the sequence length was not a power of 2, it can be zero padded to make it so [1 mark].

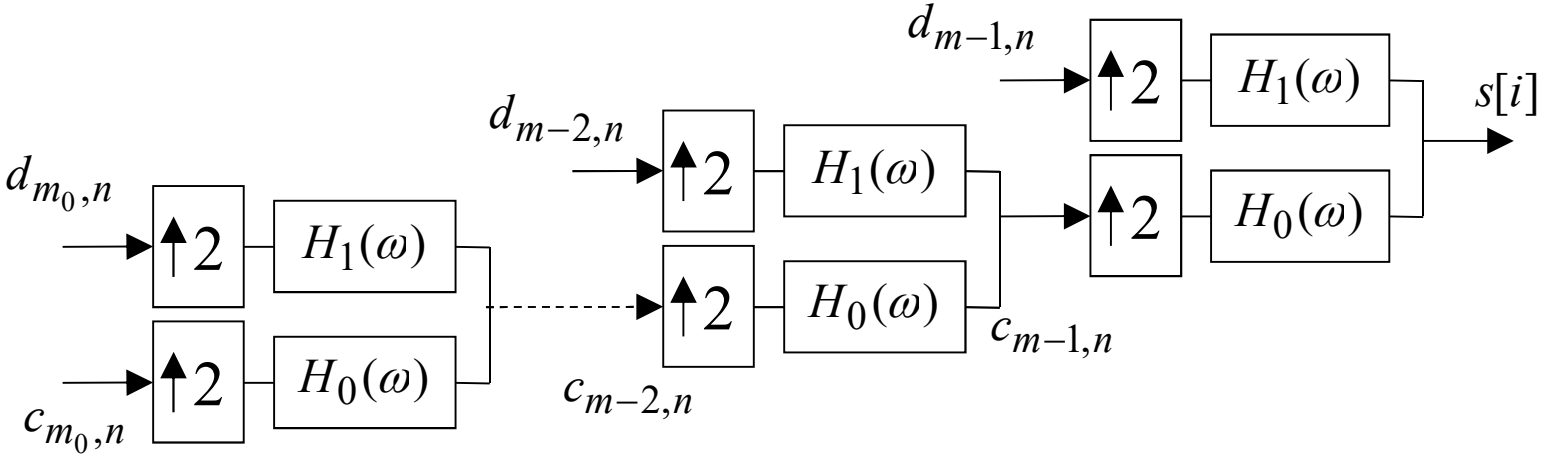
What condition is required to ensure that the original signal can be synthesized by reversing the process?

If the wavelet functions are orthogonal, then the same filters can be used as in the analysis [1 mark]. If the wavelet functions are not orthogonal, then filters that are biorthogonal with the analysis filters must be used for synthesis [1 mark].

Signal recovery filterbank

Can also compute high-res coeffs from low-res coeffs :

$$c_{m,n} = \sqrt{2}\left(\sum_i h_0[n-2i] c_{m-1,i} + \sum_i h_1[n-2i] d_{m-1,i} \right)$$



(For proof, see e.g. Qian)

So – don't need scaling functions or wavelets, just filter banks!

For Haar Wavelet Transform

Remember for the normalised Haar Functions:

$$\varphi = \frac{1}{\sqrt{2}}[1 \quad 1]$$

$$\psi = \frac{1}{\sqrt{2}}[1 \quad -1]$$

Using Recursive Equation: $\sqrt{2}(\ldots\ldots\ldots)$

So call the Haar Filters: $h_0 = \left[\frac{1}{2} \quad \frac{1}{2}\right]$, $h_1 = \left[\frac{1}{2} \quad -\frac{1}{2}\right]$

h_0: low-pass filter. Simply averaging adjacent entries of its input.

h_1: high-pass filter. computing the difference adjacent entries of its input.

Wavelet Transforms from Filter Banks

- We saw in “Wavelet Transforms from Filter Banks” that we can perform wavelet transforms by using low and high pass filters to calculate the coefficients of the transform.
- The low and high pass filters are derived from the wavelet function we want to use.
- The low pass filter is used to calculate the coefficients of the approximate output sequence (that is, the trend of the input signal) and the high pass filter to calculate the coefficients of the fine details.
- We used Haar functions as an example.
- However we know that we can obtain the output of a transform by using a transform matrix, for example as we saw with the DCT.

Wavelet Filterbanks are used for Multiresolution Analysis. Briefly explain what is meant by wavelet packets and explain two advantages of them.

Wavelet packets are a generalization of wavelets. They are a family of orthonormal bases for discrete functions of any space of order N [1 mark]. In the wavelet filterbank for dyadic wavelet decomposition only the low pass filtered subband is decomposed, but the wavelet packet allows decomposition to be carried out using any combination of filters [1 mark].

Advantages:
Flexibility [1 mark]. the best wavelet basis to satisfy any requirement can be used [1 mark]
Low computation requirement [1 mark] because the decomposition can be optimized using fast filterbanks.

Karhunen Loève Transform (KLT) – Procedures

KLT

$$\mathbf{X} = [\vec{x}_0, \vec{x}_1, \ldots \vec{x}_{N-1}]$$

1. Find mean vector for input data $E(\mathbf{X}) = \frac{1}{N}\sum_{i=0}^{N-1} \vec{x}_i$ mean(x) mean(y)
2. Find covariance matrix $\mathbf{R_{XX}} = \frac{1}{N-1}\sum_{i=0}^{N-1} (\vec{x}_i - E(\vec{x}))(\vec{x}_i - E(\vec{x}))^T$ cov(x,y) =A
3. Find eigenvalues of the covariance matrix $|\mathbf{R_{XX}} - \lambda \mathbf{I}| = 0$
4. Find eigenvectors of the covariance matrix $(\mathbf{R_{XX}} - \lambda_i \mathbf{I})\vec{\varphi}_i = 0$

} [V,D]=eig(A)

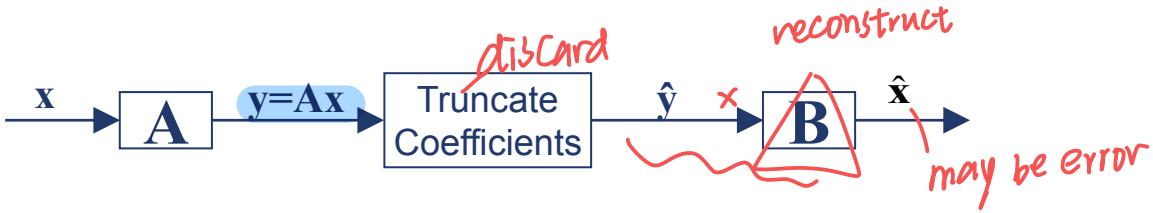
V: 特征向量 (列)
D: の (对角)

5. Normalise the eigenvectors $\vec{\varphi}^*_i = \frac{\vec{\varphi}_i}{|\vec{\varphi}_i|}$ so that $\langle \vec{\varphi}_i, \vec{\varphi}_i \rangle = 1$
6. Transform the input $\mathbf{Y} = \varphi^T \mathbf{X}$, where $\varphi^T = [\vec{\varphi}^*_1, \vec{\varphi}^*_2, \ldots]$ V'*X
7. Set the last row of the transformed data to 0
8. Invert the transform

Linear Transform Coding

- General procedures of linear transform coding (in words):
  1. **Divide image (or signal) into P blocks of N pixels (samples).**
     - The k-th block is now an $N-$dimensional vector:
       $$\mathbf{x}_k = \left(x_{1,k},\ x_{2,k},\ \ldots,\ x_{N,k}\right)^T$$
     - The image (signal) is now a sequence of vectors $\{\mathbf{x}_k\}$.
  2. **Transform $\mathbf{x}_k$ by multiplying by a linear matrix**
     $$\mathbf{y}_k = \mathbf{A}\mathbf{x}_k$$
     forward transform matrix
  3. **Transmit the first M coefficients** $\hat{\mathbf{y}}_k = \left(y_{1,k}, \ldots, y_{M,k}\right)^T$, discarding the remaining $N-M$ coeffs $y_{M+1,k} \cdots y_{N,k}$
     less significant value
  4. **Reconstruct the image block** using another matrix
     $$\hat{\mathbf{x}}_k = \mathbf{B}\hat{\mathbf{y}}_k$$
     (multiplying)
     inverse transform matrix
  5. Repeat steps 2-4 for each image block to transmit the whole image

Linear Transform Coding



- The transmission error is calculated as
  - $J = E(|\mathbf{x} - \hat{\mathbf{x}}|^2)$
  - Mean squared error (MSE)
  - $E(v)$ is the expected value (mean) of v

The KLT minimizes the mean square error J, between x and x̂ by decorrelating the input data and discarding the set with the least variance.

- We want to choose A (forward transform matrix) and B (inverse transform matrix) to minimize J

minimize error

- Easy case: if we keep all the coefficients, we have
  - $\hat{\mathbf{y}} = \mathbf{y}$, hence $\hat{\mathbf{x}} = \mathbf{B}\hat{\mathbf{y}} = \mathbf{B}\mathbf{y} = \mathbf{B}\mathbf{A}\mathbf{x}$
  - Therefore, $\hat{\mathbf{x}} = \mathbf{x}$ (zero error) if $\mathbf{B}\mathbf{A} = \mathbf{I}$, i.e., $\mathbf{B} = \mathbf{A}^{-1}$

Principal Component Analysis (PCA) – The basis for KLT

- Multivariate procedure
- Main use of PCA is to reduce dimensionality of a data set while retaining as much information as is possible.
- Finds a projection of the observations onto orthogonal axes contained in the space defined by the original variables.
- Correlated variables transformed into uncorrelated variables, known as principal components
  - Ordered by reducing variability.
  - Uncorrelated variables are linear combinations of original variables
- Computes compact, optimal description of data set.
- Rotates data so that maximum variabilities projected onto the axes
- Rotation of existing axes to new positions in the space defined by the original variables.

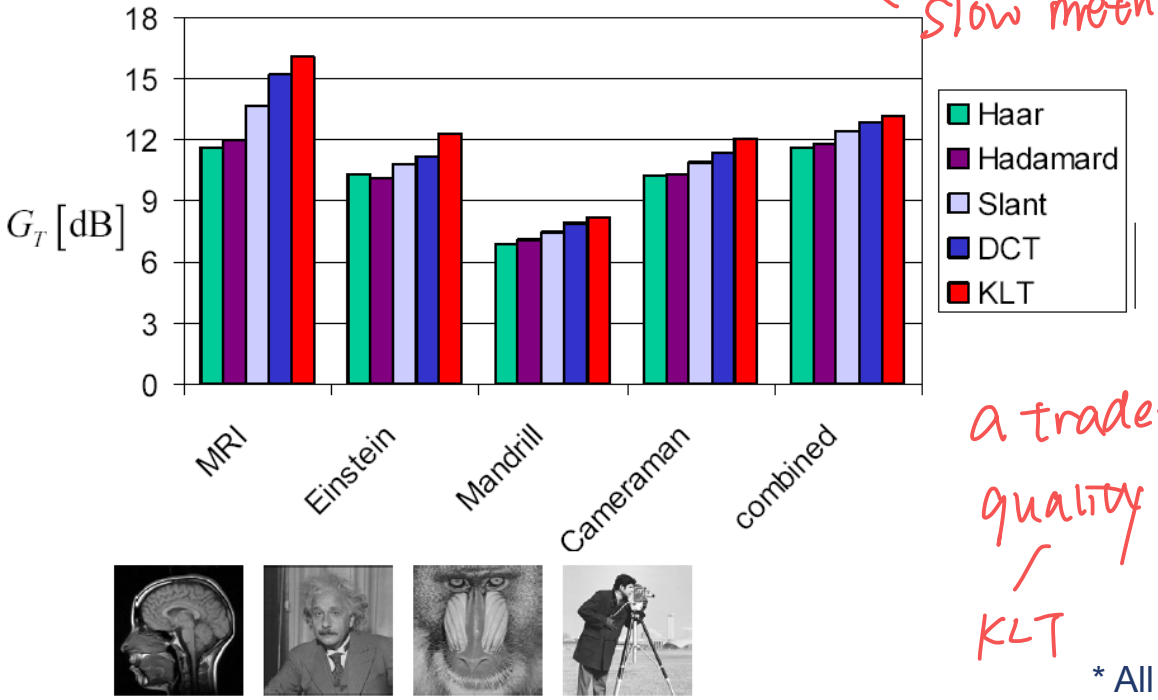
Karhunen Loève Transform (KLT)

- The generalized transform method for both discrete and continuous data
  - PCA is the discrete version of KLT
- Basis functions of KLT are eigenvectors of the covariance matrix R_{xx} of the input signal x
  - Values of the basis vectors (eigenvectors) depend on the data set, i.e., basis vectors are not fixed.
- KLT yields uncorrelated transform coefficients.
- Optimal linear transform for keeping the subspace that has largest variance
  - Achieves optimum energy concentration.
  - KLT maximizes coding gain (best "quality" of compressed data for a given level of compression)

The KLT use PCA to statiscally decouple multidimentional data by computing covariance matrix. The eigenvector of this matrix are then the basis vectors for transforming the data. The transformed data on the least principal component can be discarded.

Karhunen Loève Transform (KLT) – Coding Gain

Slow method



a tradeoff quality and efficiency

KLT | DCT

* All using 8x8 transform

Karhunen Loève Transform (KLT) - Procedures

$$\mathbf{X} = [\vec{x}_0, \vec{x}_1, \ldots \vec{x}_{N-1}]$$

To compress the data (Method 1)

1. Set last row vector(s) of $\mathbf{Y}$ to 0

$$\mathbf{Y} \rightarrow \mathbf{Y}'$$

2. Apply inverse transform

$$\mathbf{X}' = \varphi \mathbf{Y}'$$

φ: eigenvectors

$\mathbf{X}'$: estimation of original matrices

Karhunen Loève Transform (KLT) - Procedures

$$\mathbf{X} = [\vec{x}_0, \vec{x}_1, \ldots \vec{x}_{N-1}]$$

To compress the data (Method 2)

1. Set last column vector(s) of φ to 0

   least variation → less important → reduce dimension

   $\varphi \rightarrow \varphi'$

2. Apply transformation to $\mathbf{X}$

   $\mathbf{Y}' = \varphi'^T \mathbf{X}$

   Achieve at the same $\mathbf{Y}'$ as in method 1

3. Apply inverse transform

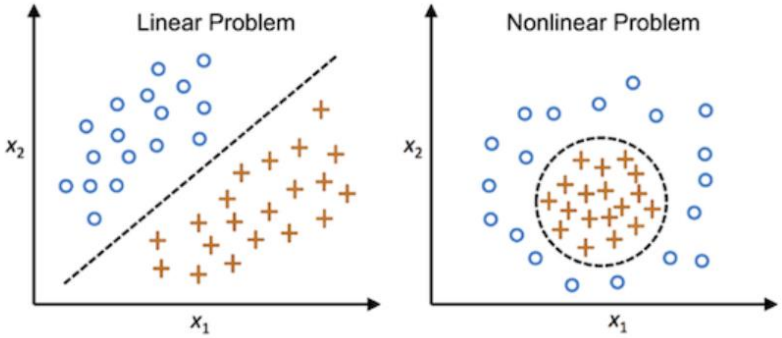
   $\mathbf{X}' = \varphi' \mathbf{Y}'$

Drawbacks of KLT

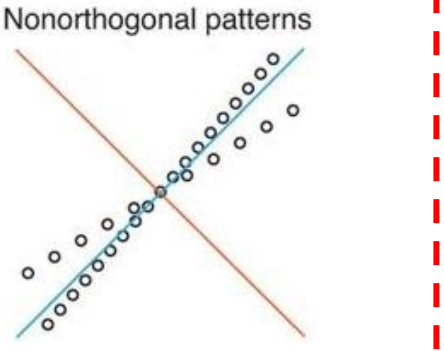
- Examples where KLT may not work well:

① 复杂 ② may not work well if non-linear / non-orthogonal

Non-linear Pattern



Non-orthogonal Pattern



[1] "Introduction to Kernal PCA with Python", *Python and R Tips*, https://cmdlinetips.com/2021/02/kernal-pca-with-python/#:~:text=Principal%20Component%20Analysis%20is%20one%20of%20the%20bread,handle%20non-linear%20data%20and%20make%20it%20linearly%20separable.
[2] "Principal Component Analysis", *Devopedia*, https://devopedia.org/principal-component-analysis

Drawbacks of KLT

KLT is theoretically optimal (in the MSE sense). (KLT maximises the coding gain, i.e. maximises the SNR after a given level of compression.)

BUT, it has practical difficulties:

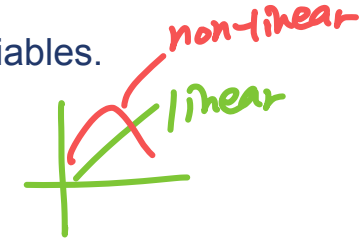
- Estimate of correlation can be unwieldy
- Solution of eigenvector decomposition is computationally intensive (i.e. slow)
- Calculation of forward and inverse transforms is $O(MN)$ for each image block
- Transmission of data-dependent basis $\mathbf{A}$ is required
- The technique is linear, therefore any non-linear correlation between variables will not be captured.

In comparison, turns out that DCT is fixed, and

- a good approximation to KLT for typical images,
- needs no eigenvalue decomposition, and
- transform is $O(N \log N)$.

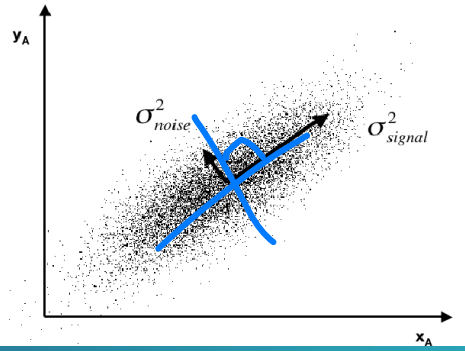
Some further points on KLT

- KLT is not suitable for low-dimensional datasets that contains a small number of variables.
  - In image processing, each pixel is counted as a variable. A typical image contains hundreds/thousands of variable.



- KLT is not suitable for non-linear datasets because the nonlinearity cannot be explained by the orthogonal principal components. Hence, all principal components will carry large weights. Compression will be very lossy.

- KLT can be designed for image compression, facial recognition, financial analysis, denoising, etc…



Summary

- Practical applications of KLT (non-assessed)
  1. Image compression
     - By keeping the top N eigenvectors that explain the majority of variance in the image
     - In the MNIST data, the number of eigenvectors can be reduced from 784 to 331 at the cost of 1% reduction in explained variance
  2. Facial recognition
     - Eigenvectors can be visualized as eigenfaces
     - Each person is associated to a unique set of eigenvalues
     - Facial recognition is achieved by comparing the set of eigenvalues in the image to the dataset

Wigner-Ville Distribution – Instantaneous power spectrum

Recall the power spectrum :

$$P(\omega) = |S(\omega)|^2 = \int_{-\infty}^{\infty} R(\tau) e^{-j\omega\tau} d\tau$$

Power spectrum of a signal is the Fourier Transform of its autocorrelation function

where $R(\tau)$ is the autocorrelation function (acf)

$$R(\tau) = \int_{-\infty}^{\infty} s(t)s^*(t-\tau)dt = \int_{-\infty}^{\infty} s(t+\tau/2)s^*(t-\tau/2)dt$$

What happens if we use *instantaneous* autocorrelation :

$$R(t,\tau) = s(t+\tau/2)s^*(t-\tau/2)$$

instead of $R(\tau) = \int_{-\infty}^{\infty} R(t,\tau)dt$? We get :

The WVD is the Fourier Transform of the instantaneous autocorrelation function

$$WVD_s(t,\omega) = \int_{-\infty}^{\infty} s(t+\tau/2)s^*(t-\tau/2)e^{-j\omega\tau} d\tau$$

which is the *Wigner - Ville Distribution* (WVD).

Wigner-Ville Distribution - Background

- So far we have looked at transforms that compute the correlation between a signal and basis functions that are functions of time and frequency (or of scale and translation). The time-frequency resolution is determined by the basis functions
  - E.g., STFT, Wavelet Transform

- An alternative approach is to compute directly:
  - ❑ Time-frequency energy density- signal's energy density in both time and frequency
    - In contrast to power spectrum: energy in frequency only
  - ❑ An example of this is the Wigner-Ville distribution

Wigner-Ville Distribution

- The Wigner-Ville Distribution is not a Windowed Transform in a similar way to the STFT

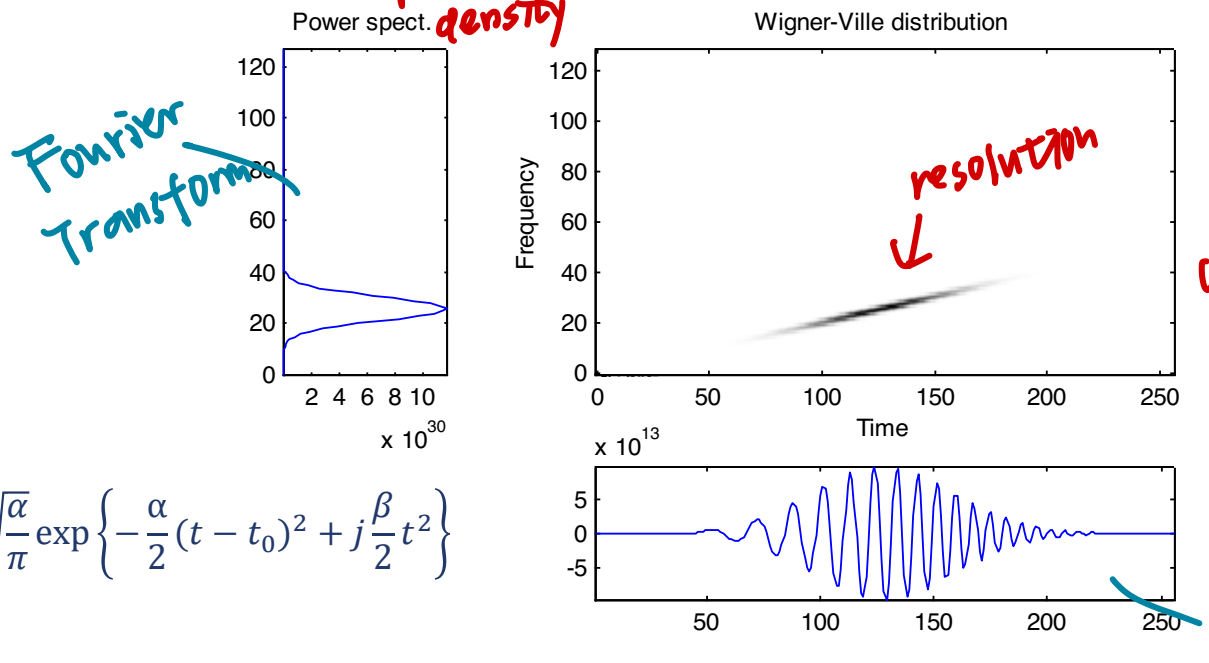
- However, the shifted version of the same signal in the autocorrelation function could be considered as a window.

- The WVD compares the information in the signal with its own information at other times and frequencies.

- The WVD has some interesting and useful properties.

Wigner-Ville Distribution

- Illustration: chirplet



$$s(t) = \sqrt[4]{\frac{\alpha}{\pi}} \exp\left\{-\frac{\alpha}{2}(t-t_0)^2 + j\frac{\beta}{2}t^2\right\}$$

With the aid of relevant diagrams, explain what is meant by the term “chirplet”.

A chirp signal is a signal whose frequency increases or decreases with time (l mark). The frequency may change linearly with time or may not be linear [l mark]. A chirplet is a short duration “windowed” chirp [1 mark]. Because the chirplet is limited in time, it has a limited frequency range so is bounded in both time and frequency [l mark].

Wigner-Ville Distribution

➢ WVD of multiple signals: Cross-terms

- Wigner-Ville Distribution has many useful properties, and better resolution than STFT spectrogram. BUT
- Applications are limited due to *cross-term interference.* Cons

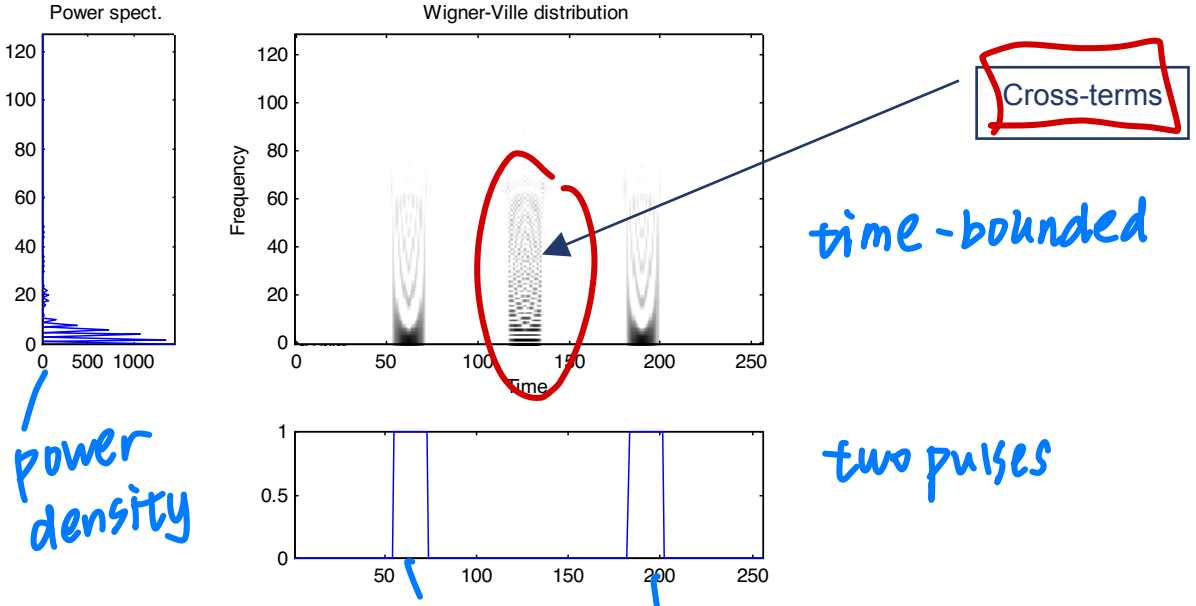
- Consider composite signal $s(t) = s_1(t) + s_2(t)$. Then
  $WVD_s(t,\omega) = WVD_{s_1}(t,\omega) + WVD_{s_2}(t,\omega) + 2\,\mathrm{Re}\{WVD_{s_1,s_2}(t,\omega)\}$
  i.e. not only the sum of WVDs, but also the cross−term $WVD_{s_1,s_2}(t,\omega)$

- Also, cross−term is included at double the magnitude of the auto−terms, so often obscures useful patterns.

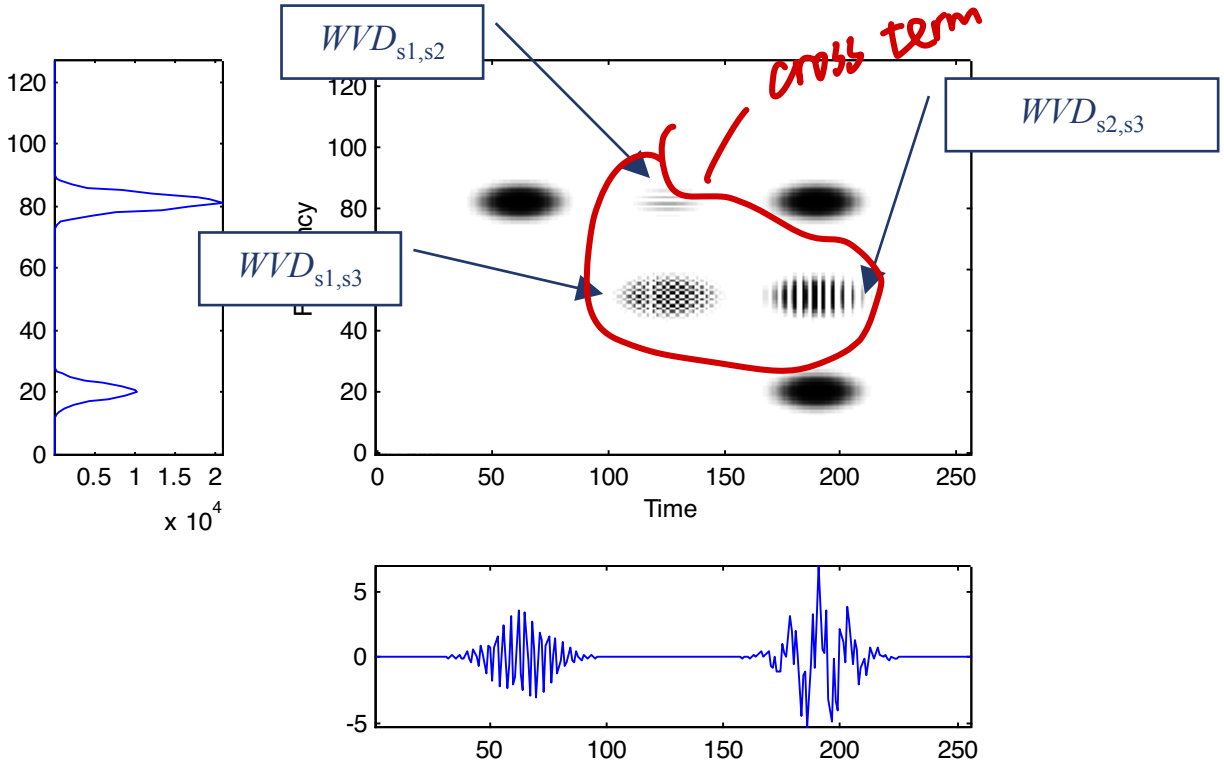
Wigner-Ville Distribution

- Example of cross-terms
  - WVD gives cross-terms for all but simplest signals, e.g.:

spectrogram



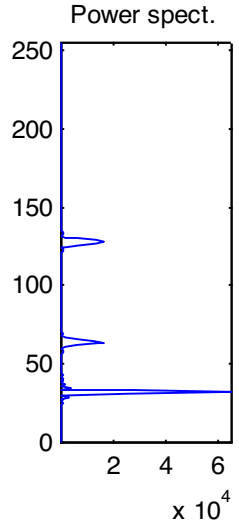
Wigner-Ville Distribution

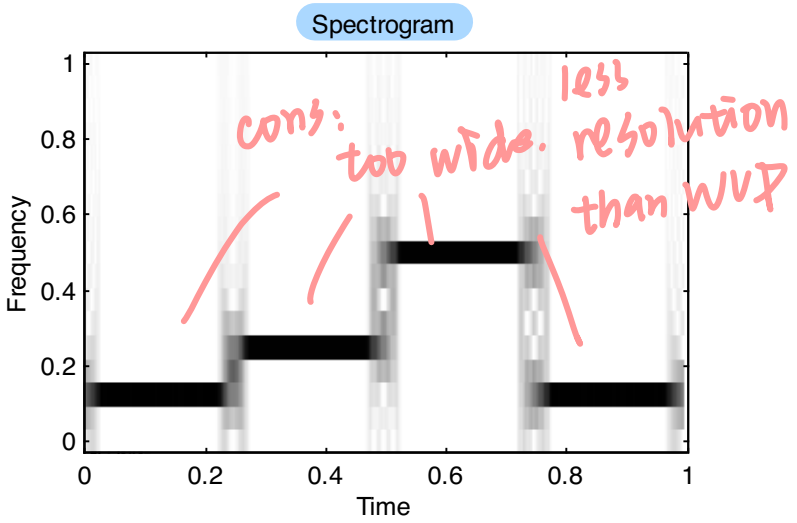


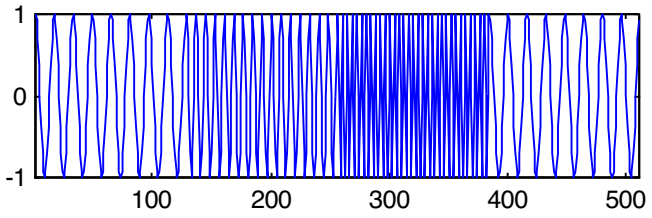
Wigner-Ville Distribution

- STFT spectrogram

pros: no cross term. more clear.







Less resolution,
but no cross-terms

Wigner-Ville Distribution - Smoothing

- So, there is a trade-off between the beneficial properties of the WVD and the interference caused by the cross terms.

way 1.

- To try to reduce the interference problem, we could use a windowed version of the WVD, called the Pseudo-WVD (PWVD):

introduce window function

$$PWVD_s(t, \omega) = \frac{1}{2\pi} \int s\left(t + \frac{\tau}{2}\right) s^*\left(t - \frac{\tau}{2}\right) h(t) e^{-j\omega t} d\tau$$

h(t) is the window function.

way 2.

- This is equivalent to frequency smoothing (low-pass filtering) the WVD

Wigner-Ville Distribution

- Since the cross−terms WVD are usually strongly oscillating, try removing them by using 2D low−pass filtering, to give a "smoothed Wigner−Ville distribution" (SWVD):

$$SWVD_s(t,\omega) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \varphi(x,y) WVD_s(t-x,\omega-y)dx\,dy$$

where $\varphi(x,y)$ is a 2D low−pass filter.

- Example: 2D Gaussian

$$\varphi(x,y) = e^{-\alpha t^2 - \beta\omega^2} \qquad \alpha,\beta > 0$$

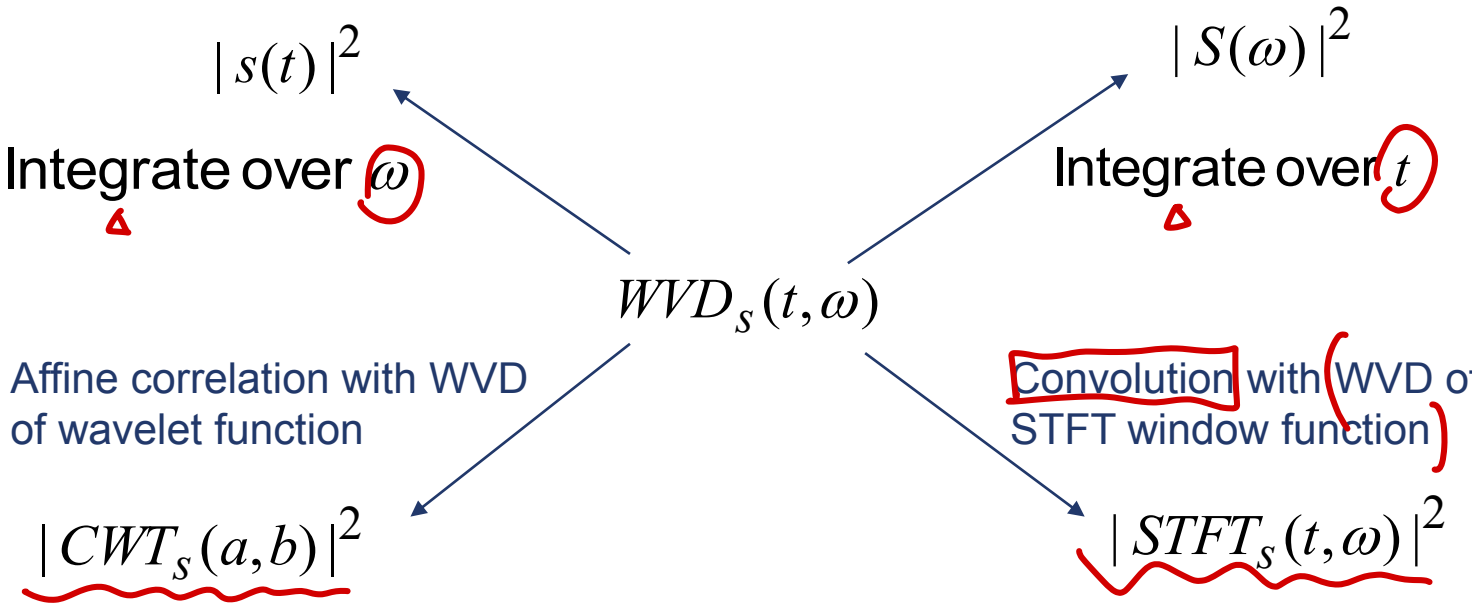
- We have a trade−off:

more smoothing → less cross−terms, BUT

more smoothing → more reduced resolution

i.e. not as good

Wigner-Ville Distribution - From WVD to …?



Both the STFT spectrogram and the WT scalogram are smoothed versions of the WVD, explaining why the WVD has the best time-frequency resolution. compared to WT. + STFT.
It is still constrained by the Uncertainty Principle.

Wigner-Ville Distribution

➢ Cross-WVD vs auto-WVD

Since we can define a cross-correlation,

we can also define a cross-Wigner-Ville distribution:

$$WVD_{s,g}(t,\omega) = \int_{-\infty}^{\infty} s(t+\tau/2)g^*(t-\tau/2)e^{-j\omega\tau}d\tau$$

Taking complex conjugates we find

$$WVD_{s,g}(t,\omega) = WVD *_{g,s} (t,\omega)$$

So for the usual WVD (auto-WVD) we have

$$WVD_s(t,\omega) = WVD_{s,s}(t,\omega) = WVD *_s (t,\omega)$$

so the auto-WVD is always real.

Wigner-Ville Distribution – How does it work?

- Power spectrum of a signal is the Fourier Transform of its auto-correlation function

$$P(\omega) = \int_{-\infty}^{\infty} R(\tau)\, e^{-j\omega t} d\tau \qquad R(\tau) = \int s(t)s(t+\tau)dt$$

- Auto-/Cross-WVD is the Fourier Transform of the instantaneous auto-/cross-correlation function

Auto-WVD: $$WVD_s(t,\omega) = \int_{-\infty}^{\infty} R(t,\tau)\, e^{-j\omega t} d\tau \qquad R(t,\tau) = s(t+\tau/2)s^*(t-\tau/2)$$

Cross-WVD: $$WVD_{s,g}(t,\omega) = \int_{-\infty}^{\infty} R_{s,g}(t,\tau)\, e^{-j\omega t} d\tau \qquad R_{s,g}(t,\tau) = s\left(t+\frac{\tau}{2}\right) g^*(t-\tau/2)$$

offset conjugate 两个信号

Wigner-Ville Distribution – Properties

- If $s(t) = s_1(t) + s_2(t)$. Then

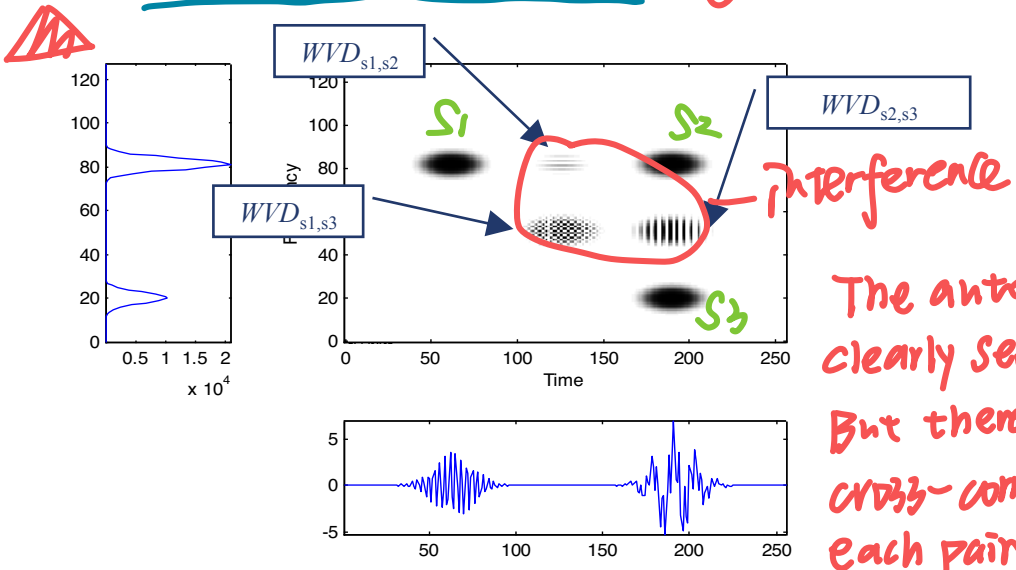
$$WVD_s(t,\omega) = WVD_{s_1}(t,\omega) + WVD_{s_2}(t,\omega) + 2\,\mathrm{Re}\{WVD_{s_1,s_2}(t,\omega)\}$$

higher magnitude (double)

Cross-term

→ So often obscure useful patterns

auto-terms



The auto correlation terms are clearly seen in the spectrogram. But there are also three cross-correlation terms, one for each pair of the components in the signal.

Wigner-Ville Distribution – Properties

- For any real or complex signal, its auto-WVD is always real-valued
  - See p10 for a simple proof
  - The cross-WVD between two signals is not always real

- WVD is invariant to time and frequency shifts
  - Time invariant: $WVD_{s_0}(t,\omega) = WVD_s(t - t_0, \omega)$
  - Frequency invariant: $WVD_{s_1}(t,\omega) = WVD_s(t, \omega - \omega_1)$
  - Both follows immediately from the equation

Wigner-Ville Distribution – Properties

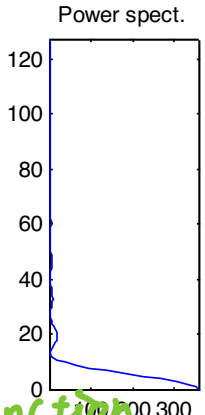
- Time marginal condition
  - $\frac{1}{2\pi}\int_{-\infty}^{\infty} WVD_s(t,\omega)d\omega = |s(t)|^2$
  - Integral over frequency of WVD is the signal power density at time t

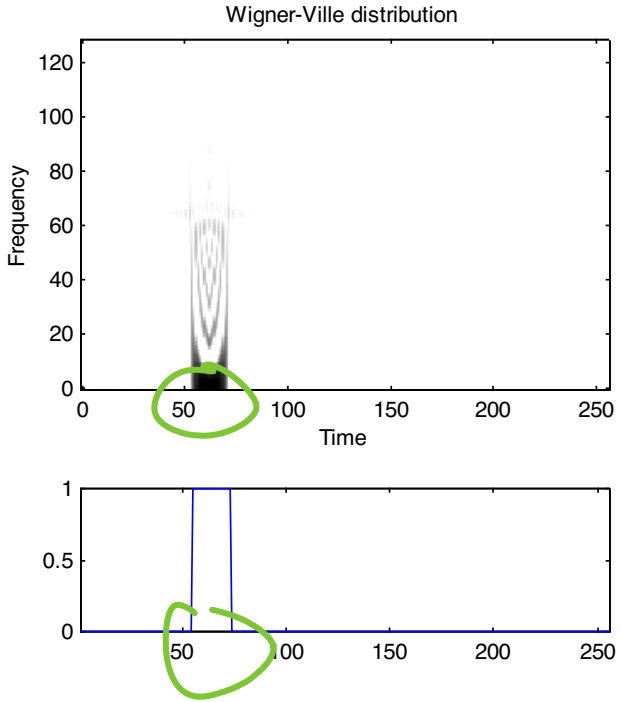
- Frequency marginal condition
  - $\int_{-\infty}^{\infty} WVD_s(t,\omega)dt = |S(\omega)|^2$
  - Integral over time of WVD is the power spectral density

Wigner-Ville Distribution – Properties

- WVD has the exact same time interval as the original time-limited signal
- WVD has the exact same frequency interval as the original frequency-band limited signal

pros than window function





Wigner-Ville Distribution - Pros

- In contrast to Fourier transform, WVD analyzes the signal from both time and frequency prospective.
  - Hence, WVD is beneficial when the signal is non-stationary

- WVD provides the highest possible time-frequency resolution which is mathematically possible within the limitations of the uncertainty principle

apply to all txf technique

Wigner-Ville Distribution - Con

- If a signal is composed of several signal components, WVD will result in large cross-terms between every pair of signal components
  - 2 components ⇒ 1 cross-term
  - 3 components ⇒ 3 cross-terms
  - 4 components ⇒ 6 cross-terms
  - …
  - N components ⇒ $(N-1)!$ cross-terms
- Moreover, the magnitudes of the cross-terms are always larger than the auto-terms

More cross-terms than auto-terms!

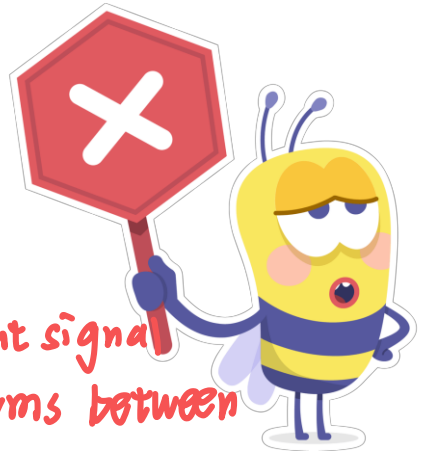
Wigner-Ville Distribution - Con

消除 interference 的办法

- The cross-terms can be suppressed by smoothing techniques
  - E.g., use a windowed version of the WVD, called the Pseudo-WVD (PWVD)
  - Similar to frequency smoothing or low-pass filtering

⇕ trade-off

- BUT, smoothing reduces the time-frequency resolution



产生 Interference 原因:

The WVD uses a autocorrelation function. However if the input signal is a composite signal then there will be cross-correlation terms between the component parts of the signal.

Wigner-Ville Distribution - Summary

- **Kind of Decomposition**

  Time-Frequency
- **Analyzing Function**

  Uses the signal itself. Motivated by time-frequency energy density (c.f. a probability density).
- **Variable**

  Time and Frequency. Has high resolution in both time and frequency.
- **Suited for**

  more complex → more cross terms

  Simple (i.e. not composite) signals (non-stationary), e.g. linear chirp, gaussian pulse
- **Notes**

  not suitable to analyse composite signals

  More "complex" (composite) signals lead to undesired "cross-terms". Can be suppressed with smoothing, but lose high resolution in the process.

Transforms Method

Transforms are used for a variety of purposes. For example:

- **Fourier Transforms** are used to obtain the frequency spectrum of a function of time.
- **Cosine Transforms** are used to identify the rate-of-change (variation) of the input data.
- **Wavelet Transforms** are used to identify trends in the input data ("approximations") and to identify short-duration features or artifacts in the input data.

Where the change occurs in the signal

Discrete Fourier Transform – Summary

- Each row of the Fourier Matrix corresponds to a cosine wave and a sine wave of the same frequency.
- Even if the input data is a sequence of real numbers, the output sequence will be complex numbers
- Multiplying an input sequence by the Fourier Matrix gives an output which is the correlation between the input data and a series of cosine and sine waves of increasing frequency.
- This is equivalent to what a Fourier Series does for a periodic input waveform.
- The 2x2 and 4x4 Fourier Matrices are relatively trivial. The 8x8 is less trivial and will not be considered here.

Discrete Fourier Transform - Tutorial

8. Explain the what DFT does to the input sequence, with reference to the DFT matrix.

What the each row means

Each row of the DFT Matrix corresponds to a cosine wave and a sine wave of the same frequency. Higher rows of the DFT matrix corresponds to a higher frequency.

By multiplying the DFT matrix with the input sequence, it computes the correlation between the input data and a series of cosine and sine waves of increasing frequency.

The output of DFT approximates the frequency spectrum of the signal that we sampled from.

Discrete Cosine Transform – Summary

- The Discrete Cosine Transform correlates an input sequence of real samples of an input signal with a set of cosine waveforms of increasing frequency.
- The output is a set of real values.
- When applied to images, it transforms from the spatial domain to a rate-of-change domain.
- It is used in the production of jpeg image compression format.

input and output: both real

DCT does compress images by itself

discard low f components

Discrete Cosine Transform – Tutorial

7. Describe what DCT does to the input signal, with reference to the DCT matrix

Each row of the DCT matrix consists of samples from a cosine wave. Higher rows in the DCT matrix represents cosine waves of increasing frequencies.

When applied to the input sequence, DCT calculates the correlation between the input and the cosine waves at different frequencies.

The output of DCT describes the rate of change in the input sequence

Discrete Cosine Transform – Tutorial

9. What is main application of DCT? Describe how DCT can be employed for that application.

- The DCT is used to perform image compression to produce jpeg format.
- For this format, an image is sub-divided into 8x8 blocks of data.
- The transform is then the dot-product of the 8x8 DCT matrix with an 8-point input sequence to produce an 8-point output sequence. Repeat this step for all blocks of data.
- This is effectively correlation of the input data with a range of cosine waves of different frequency.
- Compression is performed by setting a threshold on the DCT output, where outputs below the threshold are set to 0.
- Then, inverse DCT is performed on the compressed DCT output to obtain an accurate approximation of the original image

Discrete Wavelet Transform – Summary

- We have seen that a Haar Matrix can be constructed to perform Haar Transforms directly.
- The Haar Transform is fast because the matrix contains many zero terms and is real.
- It can be used to identify frequency components in the non-stationary input signal to be analysed and the times at which they occur (feature extraction of short-duration artifacts).
- It can be used to identify trends in the input data.
- It can be used for compression by reducing or eliminating the coefficients corresponding to high frequencies in the signal and then inverting the transform.

Discrete Wavelet Transform - Tutorial

1. What are the definitions of wavelets and a wavelet family?

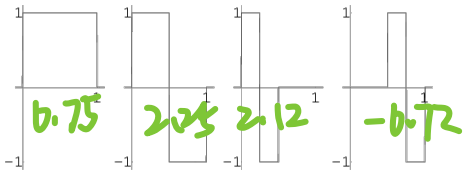
Wavelets are a class of functions which are of short duration and are oscillatory. They are used in the wavelet transform to perform time-frequency analysis.

A wavelet family is defined through the following equation:

$$\psi(t) \rightarrow \psi\left(\frac{t-b}{a}\right)$$

where $\Psi(t)$ is the basic wavelet function, b is the translation parameter, and a is the scaling parameter

Transform Methods – Exercise 1 Solution



Input: S[n] = [6, 3, -2.5, 7]

Same first output

$$\text{DFT} = \begin{bmatrix} 6.75 \\ 4.25 + 2i \\ -3.25 \\ 4.25 - 2i \end{bmatrix} \quad \text{DCT} = \begin{bmatrix} 6.75 \\ 0.835 \\ 6.25 \\ -3.845 \end{bmatrix} \quad \text{DHT} = \begin{bmatrix} 6.75 \\ 2.25 \\ 2.12 \\ -6.72 \end{bmatrix}$$

strong correlation to

large

large

$\frac{6.75}{\sqrt{N}}$

- The first output of all three methods are the same. i.e. the smoothed value, divide by $\sqrt{N}$ to obtain mean
- DFT is the only one that produces complex outputs

- Three transform methods are designed for different purposes
  - DFT – frequency spectrum as a function of time
  - DCT – rate of change in the input
  - DHT – identify short duration features

has a fast changing value at last 2 time slot.

Sharp increase in the input (∵ negative)

Notes on Comparing DFT/DCT/DHT

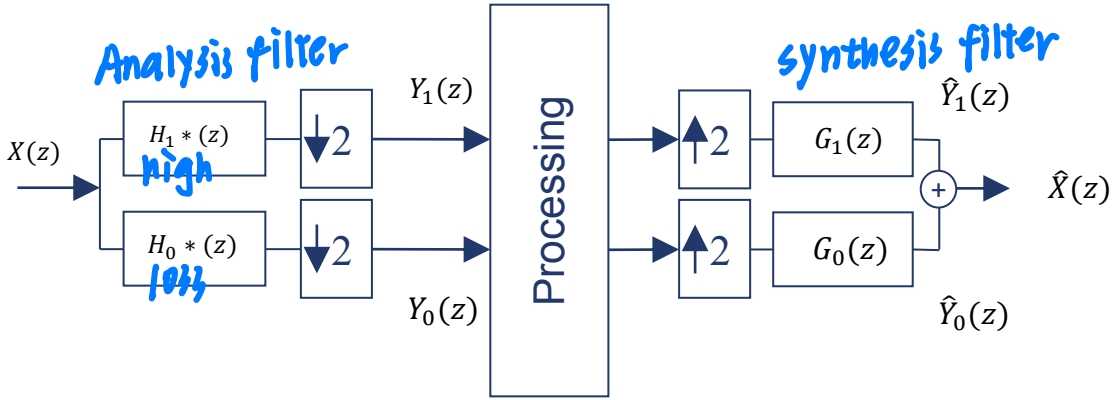
- DFT is the only transform method (among DFT,DCT,DHT) that outputs complex numbers
- DFT and DCT both assume the input to be periodic
- DCT further assumes the input to be even +real
- If the above assumption is not met, the output does not accurately represent the frequency information of the input signal that we sampled from
- The outputs of DFT and DCT on even functions are the same
- DHT (or any wavelet transform) is the only transform method (among DFT,DCT,DHT) that can perform time-frequency analysis

where the change occurs } frequency

Z-Transform and Perfect Reconstruction

Perfect Reconstruction

- Analysis filters H_0, H_1 and synthesis filters G_0, G_1 may differ.



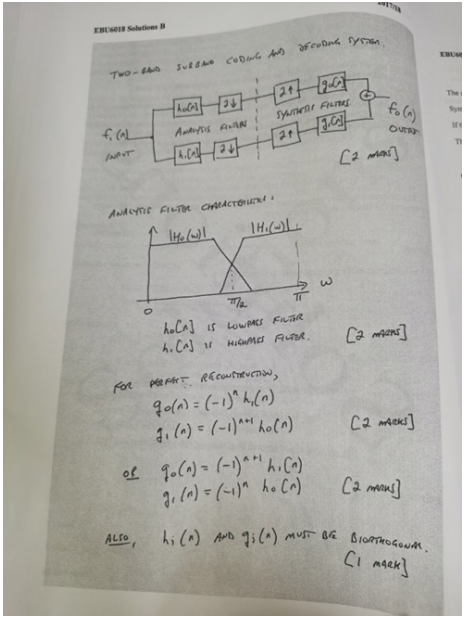
- Recall z-transform: $H(z) = \sum_{n=0}^{N} h[n] z^{-n}$
- Assuming no processing, we typically want Perfect Reconstruction (PR) - i.e. that $\hat{X}(z)$ is equal to $X(z)$
- This can be achieved with a +ve delay only, i.e. $\hat{X}(z) = z^{-l} X(z)$ for some $l \geq 0$.

The derivation is in the next slide

positive delay

With reference to a two-band subband coding and decoding system, explain the relationship between the analysis and synthesis filters.

The analysis filterbank is used to split the input sequence into two half-length sequences [1 mark]
Synthesis filterbanks combine the high and low pass subbands to produce the output [1 mark]
If the output and input are identical, then we have used perfect reconstruction filters. [1 mark]
The synthesis filters are modulated versions of the analysis filters [1 mark].
Picture:



c) With reference to the general filterbank block diagram shown in Figure xx, explain what is meant by the term "Perfect Reconstruction" and how this helps to design the filters.

[6 marks]

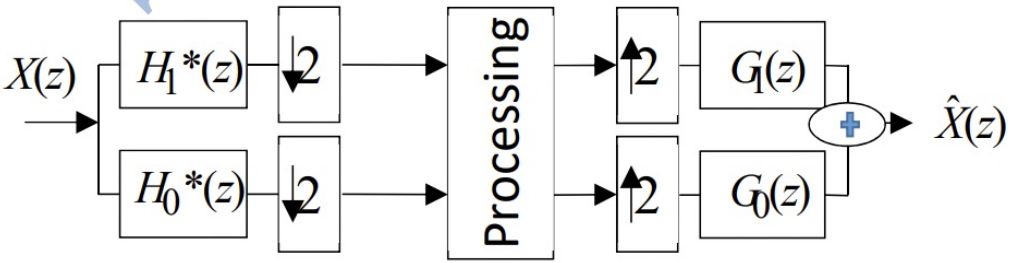


Figure Q4 c)

Answer:
c)

H and G are the analysis filters and the synthesis filters respectively. [1mark]

It is then possible to write $\hat{X}(z)$ in terms of the filters and $X(z)$ [1mark]

If we assume no processing, then perfect reconstruction means that $\hat{X}(z) = X(z)$ [1 mark]

However, there will be a pure delay between them [1 mark] and that allows us to find a relationship between the analysis filters and the synthesis filters [1 mark] and also between the analysis high pass filter and low pass filter. So if we know the coefficients of the analysis low pass filter then the other 3 filters can be determined. [1 mark]

Z-Transform and Perfect Reconstruction

- $H_0(z)$ and $G_0(z)$ are low-pass filters
  - The number of zeroes that they have and the positions of those zeroes affect the performance of the filters
  - These are FIR filters, and the position of the zeroes affects many of the filter characteristics, including phase response and orthogonality.
  - For minimum phase lag, all zeroes must lie inside the unit circle, if all zeroes lie outside the unit circle then the filter is maximum phase.
  - For example, if the factors in $P_0(z) = H_0(z)G_0(z) = (1 + z^{-1})^{2k}Q(z)$ are

$$H_0(z) = (1 + z^{-1})^2(c - z^{-1}) \text{ and } G_0(z) = (1 + z^{-1})^2\left(\frac{1}{c} - z^{-1}\right)$$

then they are orthogonal, although their phase is not linear.

Orthogonal Filter Banks – Summary

For our special case of Orthogonal Filter Banks:

- Choose $H_1(z) = -z^{-N} H_0(-z^{-1})$ ie, delayed
- $G_0(z) = H_1(-z) = z^{-N} H_0(z^{-1})$ ie, delayed
- $G_1(z) = -H_0(-z) = z^{-N} H_1(z^{-1})$ ie, delayed

$g_0(n) = (-1)^n h_1(n)$

$g_1(n) = (-1)^{n+1} h_0(n)$

or

$g_0(n) = (-1)^{n+1} h_0(n)$

$g_1(n) = (-1)^n h_0(n)$

So, if we know $H_0(z)$ then the others can be derived.

That is, the synthesis filters are time-reversed versions of the analysis filters with a delay.

Daubechies vs. Haar – Summary

- The Daubechies family is an orthogonal set of wavelet functions.
  - When k=1, we have the Haar Function
  - When k=2, we have the dbk-2 Function
- For the Haar Forward Transform, there is no overlap between successive pairs of scaling and wavelet functions
- With the dbk-2 Forward Transform there is an overlap
- The Haar high-pass filter produces a result depending on the difference between an even element and an odd element. But will not produce a result depending on the difference between an odd element and its even successor. This change will, however, be picked up in later steps.
- For the dbk-2 high-pass filter, there is an overlap so change between any two elements will be detected.

Daubechies vs. Haar – Summary

- Since the dbk-2 Wavelet Transform detects any change in the input data at every transform step, then it is more accurate in detecting changes in the input data
- However, the dbk-2 has a higher computation cost than the Haar Transform
- The trade-off depends on the application……accuracy versus speed.
- Daubechies transforms are used for detecting discontinuities and also for “self-similarity”, that is, feature extraction. They are not so useful for image processing because the phase is not linear.